

Microdosing Psychedelics for Cognitive Enhancement: A Naturalistic Exploration of User
Experiences

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Abstract

Psychedelics have recently grown in scientific interest, among this, there have been promising findings regarding its potential to treat a range of health problems (such as depression, anxiety, PTSD). Despite the growing research interest in psychedelics, there is little research investigating the effects of microdoses: tiny doses that produce sub-hallucinogenic effects. Additionally, there is little research investigating the ability of psychedelics in improving cognitive functioning. Cognitive difficulties are becoming an increasing problem worldwide with deficits arising alongside a variety of internal and external factors (e.g., psychiatric disorders, neurological damage, and general aging). Our study addresses the limitations in current research by focusing specifically on cognitive functioning and its potential mediating effects. Our study investigated the effects of psychedelic microdosing on cognitive functioning and co-occurring factors that may be mediating these effects. We hypothesised that changes in cognition will be mediated by other factors common to the psychedelic experience such as changes in mood and lifestyle. A quantitative content analysis was performed of Reddit posts collected from the r/microdosing subreddit. An exploratory survey was also distributed to collect further insight that may have been missed from the Reddit data. Results from both data sources highlight that improvements in cognition are frequently reported following microdosing. In addition, both data sources highlight prominent improvements in other domains of microdosing such as mood, social, and lifestyle changes. The findings from this study suggest microdosing acts as a cognitive catalyst within a web of personal factors, challenging the standardised approach to psychedelic medicine in favour of a personalised therapeutic design.

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Chapter One: Introduction

Psychedelics and the History of Research

The term ‘psychedelic’ was coined to categorise a variety of substances that produce mind-altering effects after consumption (see Fetty, 2024). It originates from the Greek words, psyche (for mind or soul) and deloun (for show), which loosely translates to “mind manifesting” (Fetty, 2024). This term was accepted as it captured psychedelics’ implied ability to tap into the unused potentials of the human mind. Psychedelics typically include substances such as psilocybin, LSD, DMT, mescaline, ayahuasca and ibogaine. MDMA is also often grouped with these substances, however, it does not produce the same perceptual effects as classic psychedelics (Johnson et al., 2019). This thesis focuses solely on classic psychedelics that primarily operate through 5-HT_{2A} receptor activation, therefore excluding substances like ketamine and MDMA that have different pharmacological mechanisms and risk profiles.

Psychedelics began being studied for their therapeutic effects in Western science in the 1950s, demonstrating promising results for treating disorders such as depression, anxiety and alcoholism (Hall, 2022; Krebs & Johansen, 2012; Kast, 1966, 1967 as cited in Rucker et al., 2016). However, these studies were forced to a halt in the mid-1970s due to various legal, social and economic barriers. These include, the introduction of the Controlled Substances Act, difficulty meeting new FDA (Food and Drug Administration) regulations, negative media coverage, lack of funding, and their association with drugs of abuse (Geyer, 2024; Hall, 2022; Vollenweider & Kometer, 2010). It wasn’t until the mid-1990s that psychedelics returned to the spotlight and research commenced once more in what has been referred to as the psychedelic renaissance (Sessa, 2012). Since their rebirth in science, they have continued

to show therapeutic potential for numerous psychiatric disorders, as detailed in the following section (Chao & Horton, 2021; Dawood Hristova & Pérez-Jover, 2023).

Current Understandings: Psychedelics' Therapeutic Potential

21st-century clinical studies on psychedelics have shown promising therapeutic potential for disorders such as depression, anxiety, PTSD, substance abuse, and eating disorders (Breeksema et al., 2020; Ko et al., 2023; Krebs & Johansen, 2012; Vargas et al., 2020). These clinical results have helped reshape scientific and public perceptions of psychedelics, gradually shifting their status from a stigmatised substance to a promising therapeutic agent (Eck, 2022). Building on the positive outcomes of early clinical trials, researchers are now exploring their potential to relieve a broader range of conditions, such as: traumatic brain injury, autism, cluster headaches, Obsessive Compulsive Disorder (OCD), and Attention-Deficit/Hyperactivity Disorder (ADHD) (Buot et al., 2023; Cherian et al., 2024; Haijen et al., 2022; Markopoulos et al., 2022; Shirane et al., 2023). A Comprehensive review of psychedelics' therapeutic applications across various health conditions can be found in recent systematic reviews by Cavarra et al., (2022) and Yao et al., (2024).

Microdosing: The Promising Frontier in Psychedelic Research

Despite the growing interest in the therapeutic effects of psychedelics, research has predominantly focused on macrodoses. In contrast, the practice of microdosing remains under researched. Microdosing, as defined by Kuypers and colleagues (2019), is the practice of consuming very small, sub-hallucinogenic doses of a psychedelic substance, 5-10% of a usual psychoactive dose (7–13ug of LSD or 0.1-0.4g of psilocybin (Fadiman & Korb, 2019). Though doses vary across studies, with doses as high as 26ug of LSD or 0.5g of psilocybin (Bershad et al., 2019; Cavanna et al., 2022)). These microdoses are consumed using a repeated dosing schedule (e.g., every third day as proposed by Fadiman (2011)), and taken

without the intention of experiencing typical high-dose psychedelic effects (e.g., mystical experiences and visual distortions).

Although the narrative is changing, psychedelics have historically been perceived by the public and depicted in the media as a dangerous category of drugs (see Bracco, 2019). However, they are physiologically one of the safest classes of drugs acting on the central nervous system. This is because research indicates psychedelics do not lead to addiction or dependence, they have very low overdose potential, and long-term use is not associated with increased negative mental health outcomes (Krebs & Johansen, 2013; Nichols, 2016a; Schlag et al., 2022). Despite their physiological safety, macrodoses lead to experiences that are often very intense and can put users under increased psychological risk in the event of a “bad trip” (overwhelming distress) (Amsterdam et al., 2011; Johnson et al., 2008). Microdosing can offer a solution to these psychological concerns, as sub-hallucinogenic doses allow users to circumvent the risk of adverse psychological experiences sometimes observed with higher doses.

Furthermore, another practical advantage of microdosing relates to the time commitment involved. The duration of a psychedelic trip is between four to six hours for psilocybin and up to twelve hours for LSD, which is a substantial amount of time spent intoxicated by the psychedelic effects (see Nichols, 2016b). In contrast, microdosing produces subtle effects that don't impair daily functioning, a distinction particularly relevant for therapeutic applications. If microdosing consistently produces the positive effects reported in naturalistic studies (see Petranker et al., 2022), it could provide benefits while reducing the need for expensive clinical oversight typical of full-dose psychedelic psychotherapy (Marseille et al., 2022). This is supported by anecdotal reports that highlight a preference for microdosing over high doses of psychedelics because it is less intense and

tasking, but still a tangible process with gradual changes over time (Andersson & Kjellgren, 2019).

The practice of microdosing gained significant popularity and is often attributed to the publication of James Fadiman's book *“The Psychedelic Explorers Guide”* (2011), which was the first publication to describe microdosing in detail (for more information, see Polito & Stevenson, 2019). In this book, Dr. Fadiman discusses the benefits of early psychedelic microdosing case reports. Additionally, media coverage of Silicon Valley workers who went public about their use of LSD as a method for enhancing their creative abilities and efficiency, further popularised microdosing (Glatter, 2015; Sahakian et al., 2018). Most recently, microdosing has taken off across the internet. Online discussion forums enable users to share experiences and exchange information in ways that make the practice accessible to others. This online growth is demonstrated in one study which found that 71% of microdosers learned the practice from internet sources, showing how crucial digital platforms are in spreading information on microdosing (Lea, Amada, Jungaberle, et al., 2020a).

Microdosing is often portrayed as a “multi-enhancer” in the microdosing community, that produces benefits spanning various areas of well-being (Liokaftos, 2021). These include improved emotional state (improved mood), enhanced social connections and interpersonal relationships (greater sociability and empathy), physiological improvements (alleviated headaches and better sleep), reduced symptoms associated with psychiatric disorders (reduced depression and anxiety), and cognitive enhancements (increased creativity, memory and focus)(see; Anderson et al., 2019; Miura, 2022). While these claims suggest microdosing can have positive effects on multiple aspects of well-being, it is important to note that most of the reported benefits are anecdotal, and some accounts even mention impairments in the same domains (Petranker et al., 2022). Cognitive functioning, in particular, commonly produces

contradictory findings, such that some individuals report improvements in areas such as focus and memory, while others experience impairments (Anderson, Petranker, Christopher, et al., 2019). The variability is clearly shown in a study by Cameron et al., (2020) which investigated the perceived effects of microdosing on memory and attentional focus. The authors reported that of all participants, 38.5% reported improved memory, 13.1% worsened memory, and 46.5% no effect. For attentional focus, 58.97% reported improvement, 14.74% reported worsening, and 26.28% reported no effect. This inconsistency in cognitive outcomes suggests that the effects of microdosing may be influenced by individual differences, contextual factors, and methodological variations across studies. To better understand the variation in cognitive outcomes, our study investigates the effects of psychedelic microdosing on cognitive functioning and how these effects may interact with other subjective experiences and contextual factors.

Cognitive Impairment: Increasing Prevalence and Treatment Limitations

Investigating the potential of microdosing for cognitive enhancement is becoming increasingly important, as an increasing proportion of the population suffer from cognitive difficulties due to various internal and external factors which I will outline below. For instance, executive functioning, attention, and memory are vulnerable domains often impaired in psychiatric disorders such as depression, OCD, ADHD, substance abuse, and schizophrenia (Trivedi, 2006). The World Health Organisation (2022) estimated that one in every eight people worldwide lives with a mental illness, and this prevalence is only expected to rise over time (Wu et al., 2023). Cognitive impairment can also develop as a result of neurological diseases like Alzheimer's, which causes gradual deterioration of memory and thinking skills (National Institute on Aging, 2021). Neurological damage caused by events such as a traumatic brain injury or stroke also leads to varying cognitive impairments, depending on the region damaged (Trivedi, 2006). Age-related cognitive decline is also

relatively common, with a meta-analysis finding that 5% to 41% of individuals over the age of 50 experience impairments in cognition, predominantly difficulties with memory, learning and concentration (Pais et al., 2020). Finally, environmental factors can also contribute to cognitive impairments, such as pollutants caused by increasing urbanisation and neighbourhood social environments (Luo et al., 2024).

Despite the increasing prevalence of cognitive impairments, effective treatments to combat these symptoms are lacking. For instance, current depression treatments have no established evidence of improving cognitive impairments (Dhakal & Bobrin, 2024). Individuals suffering cognitive impairments from a mild traumatic brain injury are commonly advised by clinicians to rest and wait for the symptoms to go away with time, which may take months or even years (Post-Concussion Syndrome, n.d.). Additionally, attempts to improve cognitive deficits following a stroke have been largely unsuccessful (Husain & Mehta, 2011). These examples highlight the need for research investigating microdosing as a potential cognitive enhancer.

Neurobiological Foundations of Microdosing: From Receptors to Cognition

Serotonergic Mechanisms and Neurotrophic Effects

This section explores how neurobiological research provides evidence supporting the potential use of microdosing for improving cognitive functioning. Classic psychedelics primary effects occur by directly binding to and stimulating Serotonin 2A (5-HT_{2A}) receptors, a receptor that is widely distributed in the brain and significantly involved in mediating memory and cognition (Zhang & Stackman, 2015). The activation of these 5-HT_{2A} receptors appears to drive the neuroplasticity-enhancing effects, leading to more pronounced effects in brain regions with greater 5-HT_{2A} receptor expression, such as the cortex (Beliveau et al., 2017). This 5-HT_{2A} receptor specific activity helps explain why

animal studies have found greater neuro-plasticity changes in the frontal cortex than in the hippocampus (Calder & Hasler, 2023; Jepsen et al., 2021).

Through 5-HT_{2A} receptor activation, psychedelics exert multiple neurotrophic properties and have recently been labelled a “psychoplastogen”, acknowledging their ability to produce neurostructural changes after consumption (Olson, 2018). These neurotrophic abilities are believed to occur due to their ability to increase Brain-Derived Neurotrophic Factors (BDNF) (Hutten et al., 2021). BDNF are small proteins that promote neuronal growth and synaptic plasticity (Marton et al., 2019; Song et al., 2017). The neurotrophic effects of psychedelics that have been identified in research include (1) synaptogenesis (birth of new synaptic connections), (2) neurogenesis (birth of new neurons), (3) synaptic plasticity (ability to strengthen or weaken synaptic connections), and (4) spinogenesis (birth of new dendritic spines) (de Vos et al., 2021; Ly et al., 2018).

Regional Brain Effects: Cortical vs Hippocampal Impact

These neuroplasticity mechanisms have particularly significant implications for the frontal cortex, which plays an important role in various high-level cognitive functions, including executive functioning, attention, and working memory (E. K. Miller & Cohen, 2001). Building on the 5-HT_{2A} receptor mechanisms described above, research in mice has demonstrated that psychedelic administration increases BDNF specifically in the prefrontal cortex (Zhao et al., 2024). Importantly, an increase in frontal cortex BDNF has been directly linked with improvements in prefrontal-cognitive functioning (Hwang et al., 2016).

Consistent with this literature, we hypothesise that psychedelic microdosing will enhance higher-order cognitive functions through this BDNF-mediated mechanism. Therefore, we predict to observe a significant number of reports indicating an increase in prefrontal-cognitive functioning (e.g., attention, creativity, and problem-solving).

While the cortical effects of microdosing on neuroplasticity appear relatively consistent, the impact on the hippocampus presents a more complex picture. The hippocampus is a midbrain region responsible for cognitive functions related to learning and memory (Fan et al., 2018), and relies critically on neuroplastic changes such as the formation and alterations of synapses and dendrites (Catlow et al., 2016; Gulyaeva, 2017). Unlike the changes observed in the cortex, the effects of psychedelics on neuroplasticity in the hippocampus are mixed and typically less pronounced.

This difference becomes apparent when examining conflicting evidence. For example, in a study with humans, psilocybin had the ability to strengthen hippocampal synapses, suggesting a potential positive effect on the hippocampus (Hesselgrave et al., 2021). In contrast, another human study found that psilocybin reduces hippocampal glutamate activity (the primary excitatory neurotransmitter involved in neurogenesis and synaptic plasticity), suggesting a potential inhibitory effect on the hippocampus (Mason et al., 2020). Research with pigs further complicates the picture, showing an acute decrease in 5-HT_{2A} receptor density (suggesting an inhibiting effect), yet also providing evidence of increased long-term synaptogenesis (supporting an enhanced effect) (Raval et al., 2021).

Calder and Hasler (2023) suggest that this disparity in findings is likely explained by differences in serotonin receptor distribution between brain regions. This is because the hippocampus has a greater density of 5-HT_{1A} than 5-HT_{2A} receptors, unlike the cortex where 5-HT_{2A} receptors are more prominent (Beliveau et al., 2017; Mengod et al., 2015). This key difference in receptor distribution likely accounts for the regional variation in psychedelic effects. Consistent with these findings, we hypothesise that microdosing will have a stronger effect on regions with a greater density of 5-HT_{2A} receptors, such as the cortex, while exerting less pronounced or even inhibitory effects on 5-HT_{1A} receptor-

dominant regions like the hippocampus. Therefore, we predict to observe more reports indicating an improvement in cognitive abilities pertaining to the frontal cortex (e.g., creativity, problem-solving, focus) than abilities pertaining to the hippocampus (e.g. memory and learning).

The Dosage Question: Neuroplastic Implications

Beyond regional differences in receptor distribution, dosage also plays a critical role in how psychedelics affect neuroplasticity, particularly in the hippocampus. Specifically, research with mice found that low doses of psilocybin were associated with either an increase or no effect on hippocampal neurogenesis, while higher doses were associated with reduced hippocampal neurogenesis (Catlow et al., 2013). This emphasises the importance of dosage considerations in microdosing research, as varying dosages may have a different effect on cognitive functions relating to the hippocampus. The non-linear relationship between dosage and neurogenesis implies that the cognitive effects of microdosing may follow a bell-shaped curve, where the top of the curve represents optimal cognitive enhancement. Our study aims to provide insight into the optimal dose that best predicts cognitive enhancement.

Targeting Root Causes: Cognitive Improvement Through Disorder Management

This section explores how the neurotrophic and neuroprotective effects of psychedelics might improve cognition by indirectly relieving co-occurring psychiatric and neurodegenerative disorders. These conditions share common neuro-structural abnormalities, notably the atrophy of cortical neurons, which is believed to contribute to the development of both psychiatric and neurodegenerative disorders (Christoffel et al., 2011; Duman & Aghajanian, 2012; Mouton et al., 1998). Psychedelics may improve these disorders by increasing BDNF levels and promoting neuronal growth, which is a process that research has linked to improvements in both depression and Alzheimer's disease (Brunoni et al., 2008;

Gao et al., 2022). Additionally, psychedelics demonstrate anti-inflammatory properties that further emphasise their therapeutic potential for cognitive disorders (Flanagan & Nichols, 2018). While inflammation typically serves as a protective response facilitating healing, prolonged inflammatory states can lead to tissue damage (Sartori et al., 2012). This chronic neuroinflammation plays a pathological role in Alzheimer's disease and has been linked to both cognitive decline and risk of dementia (Ahmad et al., 2022; Godbout & Johnson, 2006; Latta et al., 2015). Similarly, research has indicated that psychedelics may improve cognitive functioning deficits resulting from traumatic brain injury through their ability to reduce neuro-inflammation and promote neural re-wiring (Reichelt et al., 2023). Based on these mechanisms, we hypothesise that the cognitive enhancements observed from microdosing may be partially mediated through the alleviation of pre-existing disorders.

Literature Review

The current literature surrounding microdosing and cognition is ambiguous as it consistently reports contradictory findings. The discrepancies between studies follow a notable pattern based on their methodological approach. Studies utilising subjective data, including online reports, surveys, and interviews, typically identify positive effects spanning multiple areas of cognition. In contrast, lab-based studies that employ validated cognitive measures report mixed effects, with most finding no significant changes. This discrepancy between subjective experiences and objective measurements raises important questions about the true effect of microdosing on cognitive functioning and what factors may be influencing these outcomes. To systematically address these questions, our review of the existing literature will proceed in three parts. First, we will summarise findings from lab-based studies that administer objective cognitive assessments. Second, we will summarise studies that provide a unique perspective by combining objective measures with subjective scales. Finally, we will summarise naturalistic studies that rely on subjective reports of cognitive

changes. By examining these different methodological approaches side by side, we aim to provide a more comprehensive understanding of how different methodologies may influence the cognitive change identified. In addition, we note limitations in current research and how they should be addressed in future research.

Objective Studies on Cognitive Effects

Eight studies have investigated the effect of microdosing on cognitive functioning using objective measures, with findings providing minimal support for cognitive enhancement (see Table 1 for a summary of these studies). Only one study found evidence of cognitive improvement (Prochazkova et al., 2018), specifically in creativity, though importantly, the same study found no effect on fluid intelligence. Similarly, Cavanna et al., (2022) reported impairments in attention and inhibition, but no significant effect on creativity or cognitive flexibility. These results indicate that both improvements and impairments were specific to particular cognitive domains rather than reflecting a global cognitive change. The remaining six studies that administered objective measures of cognition demonstrated no significant effect on cognitive functioning. One noteworthy study by Yanakieva et al. (2019). found that microdosing led to longer time reproduction, indicating an altered perception of time that cannot be easily categorised as either cognitive improvement or impairment.

Collectively, these lab-based studies present a consistent pattern that shows minimal support for microdosing to enhance cognitive functioning. The individual finding of improved creativity is substantially outnumbered by studies finding no significant effect across various cognitive domains. These results suggest that microdosing may not produce reliable, objective cognitive enhancements detectable by standard laboratory measures.

Table 1:
Summary of Studies Employing Objective Cognitive Assessments of Psychedelic Microdosing

Study	Substance (Dose)	Participants (N of males; Mean age)	Num of doses	Cognition
Bershad et al., 2019	LSD (0, 6.5, 13, 26µg)	20 Healthy young adults (19-30) (8; 25 ± 3)	3 (1-week intervals)	No effect: Working memory (N-Back), cognitive functioning (DSST), & creativity (Correct RAT Scores)
Cavanna et al., 2022	Psilocybin (0.5g)	34 intending to microdose (23; 31.26 ± 4.41)	1	Impaired: Attention (Attentional blink) & inhibition (Stroop) No effect: Creativity (RAT & AUT & CPS) & Cognitive Flexibility (CFS)
Fallon, 2021			1	No effect: Working memory (Ignore/Update task)
Family et al., 2020	LSD (0, 5, 10, 20ug)	48 Healthy elderly volunteers (55-75yrs) (27; 62.92 ± 5.65)	6 doses (every 4th day)	No effect: Reaction time, Visual memory & learning, Visual Attention, Spatial working memory (All measured by CANTAB)
Yanakieva et al., 2019				Longer Time perception (Temporal reproduction task)
Forsyth et al., 2016	Ibogaine (20mg)	21 Healthy male individuals (Mean age 23.5)	1	No effect: Inhibitory control (Anti Task), switching abilities (Pro/Anti Task), attention/inhibitory control (Simon task & Flanker task), short term memory (Forward Spatial Task), working memory (Backward Spatial task & 2-back task)
Murphy et al., 2024	LSD (10ug)	80 Healthy male volunteers LSD N = 40 (37.4 ± 9.2) Placebo N = 40 (36.3 ± 7.2)	14 doses (every 3 rd day)	No effect: Creativity (AUT, CAT, RAT)
Prochazkova et al., 2018	Psilocybin Truffle (0.22g, 0.33g, 0.44g (Mean .37g))	38 Volunteers at MD event (23; 31.1 ± 11.49)	1	Improved: Creativity (PCT, AUT: fluency, flexibility, originality) No effect: Fluid intelligence (RPM), creativity (AUT: Elaboration measure)

Note. The following acronyms represent cognitive assessment measures: DSST = Digit

Symbol Substitution Task; RAT = Remote Associates Task; AUT = Alternative Uses Task;

CPS = Creative Personality Scale; CFS = Creative Flexibility Scale; CANTAB = Cambridge

Neuropsychological Test Automated Battery; CAT = Consensual Assessment Technique;

PCT = Picture Concept Task; RPM; Raven's Progressive Matrices.

Mixed Methodologies Findings

Five studies employed a mixed-methods approach combining objective cognitive assessments with validated rating scales (see Table 2 for a summary of these studies). This mixed methods approach provides a unique opportunity to compare how objective performance differs from scale-based assessments within the same participant. These studies present a complex pattern that further complicates our understanding of cognitive effects caused from microdosing. All five studies reported null findings across multiple cognitive measures, aligning with the frequent null findings from purely objective studies discussed

above. However, two of these studies also demonstrated both improvements and impairments in specific contexts, demonstrating potential dose-dependent effects. Sziegeti et al. (2021) found increased reasoning ability, but only in participants who completed the full 8-dose microdosing protocol (microdose group), while impaired short-term memory was observed exclusively in participants who completed a 4-dose protocol (half-half group). Similarly, Hutten et al. (2020) observed improved attention following both 5µg and 20µg doses of LSD, but not following 10µg doses, while working memory was impaired only at the highest 20µg dose.

The subjective scale-based assessments used in these studies produced similarly mixed findings to the objective measures. Two studies reported improvements in creativity (Murphy et al., 2023; Sziegeti et al., 2021), while three studies found no significant effects on focus and cognitive control dimensions (de Wit et al., 2022; Hutten et al., 2019; Molla et al., 2024). Interestingly, dose-dependent effects were also observed in self-reported measures, mirroring the pattern seen in objective assessments. Hutten et al. (2020) found that concentration was improved after a 10µg dose of LSD, but was impaired after a 20µg dose. Interestingly, Sziegeti et al. (2021) demonstrated that creativity improvements were only significant when participants believed they were taking a microdose, highlighting the potential role of expectancy effects.

Collectively, these studies present a complex pattern of results that further complicates our understanding of the cognitive effects of microdosing. These mixed findings demonstrated in objective measures and validated scales indicate that cognitive changes are rarely observed objectively, and may be dose dependent. Overall, these studies provide limited support for reliable cognitive enhancement from microdosing, with most measures showing null or highly context-dependent effects.

Table 2:

Summary of Studies Combining Objective Cognitive Assessments and Validated Self-Report Measures of Psychedelic Microdosing

Study	Substance (Dose)	Participants (N of males; Mean age)	Num of doses	Cognition
de Wit et al., 2022	LSD (0, 13, 26ug)	56 Healthy adults (Aged 18-35) (37; 25.63 ± 4.36)	4 doses, Every 3-4 days	<i>No effect:</i> Working memory (N-back task), Working Memory & attention (DSST), impaired control & cognition (5D-ASC*) Self-reported greater performance in high dose group (compared to other groups)
Hutten et al., 2020	LSD (0, 5, 10, 20ug)	24 healthy individuals (12; 22.8 ± 3.0)	1	<i>Improved:</i> Attention (PVT) & Concentration (10ug; VAS*) <i>Impaired:</i> Concentration (20ug; VAS**), productivity (20ug; VAS*), & increased confusion (20ug; POMS**), working memory & attention (DSST) <i>No effect:</i> Cognitive control (CCT)
Molla et al., 2024	LSD (26ug)	39 with mild depressed mood (age 18–35 years) Low BDI: N = 19 (10; 25.5 ± 3.6) High BDI: N = 20 (11; 26.9 ± 3.6)	1	<i>No effect:</i> Creativity (DAT, FF, RAT, AUT) <i>Impaired:</i> Control and cognition (5D-ASC*)
Murphy et al., 2023	LSD (10ug)	80 Healthy male volunteers LSD N = 40 (37.4 ± 9.2) Placebo N = 40 (36.3 ± 7.2)	14 doses every 3 rd day	<i>Increased:</i> Creativity (VAS*) <i>No effect:</i> Fluid intelligence (Composite score of NIH cognitive battery), focus (VAS*)
Szigeti et al., 2021	Could choose own substance and dose LSD (N=147) (Mean: 13ug) LSD analogue (N=33) (Mean: 13ug) Psilocybin (N=57) (Mean: 0.2g)	191 healthy individuals with intention to microdose (138; 33.5 ± 9.4)	2 doses a week (0/4/8 total – depending on condition)	Cognitive performance score (combined all tests): <i>No effect:</i> MD group <i>Decreased:</i> HH group. Individual tests: <i>Increased:</i> Reasoning ability (Rotations & odd one out (Only MD group)), & Creativity** (VAS*) <i>Decreased:</i> Short term memory (Spatial span & paired associates (Only HH group)) <i>No effect:</i> Focus (VAS*)

Note. The following acronyms represent objective cognitive assessment measures: DSST =

Digit Symbol Substitution Task; PVT = Psychomotor Vigilance Task; CCT = Cognitive

Control Task; RAT = Remote Associates Task; AUT = Alternative Uses Task; FF = Forward

Flow; DAT = Divergent Associates Task. * Represents scale based cognitive assessments and

the following acronyms represent: 5D-ASC = 5 Dimensional Altered States of

Consciousness; VAS = Visual Analogue Scale; POMS = Profile of Mood States. **

highlights that creativity was only significant when participants believe they are taking a MD

Subjective Reports of Cognitive Effects

16 naturalistic studies have investigated the subjective effects of microdosing on cognition through various methodological approaches: surveys, interviews, and qualitative content analyses (see Tables 3-5 for summaries of these studies). Unlike laboratory studies, these naturalistic investigations consistently identify cognitive benefits across multiple domains. While some studies noted challenges, the overall pattern in subjective reports

strongly favours cognitive enhancement. This contrast with laboratory findings suggest a significant disconnect between measured cognitive performance and lived experience.

Eight survey studies, collectively representing thousands of participants, demonstrate that improvements were the most frequently reported cognitive change, although impairments were also frequently identified (Table 3 summarises these survey studies). Multiple cognitive domains showed a similar pattern of enhancement accompanied by less frequent reports of impairment. Improvements in attention/focus (14.8-59% across studies) and memory/cognitive clarity (29-45%) were commonly reported, while impairments in these same domains were noted at lower rates (attention: 8-15%; memory/clarity: 8-13%). Creativity enhancements were consistently reported in high proportions across studies (Anderson et al., 2019: 12.9%; Lea et al., 2020: 54.1%; Petranker et al., 2022: 47-49%), making it the most commonly reported cognitive domain to benefit from microdosing with no reports of impairment. These survey findings highlight that while cognitive enhancements are reported at much higher rates, effects vary significantly between individuals, with some experiencing impairments in the same domains where others report benefits.

Table 3:

Summary of Studies utilising a Survey Design to Investigate the Cognitive Effects of Psychedelic Microdosing

Study	Sample	Microdosing practices	Effect on cognition
Anderson et al., 2019	N = 278 (Experience with microdosing) Mean age: 27.8, Male: 85.3%	Psilocybin-only (18%) LSD-only (70.1%) Both (11.9%)	Benefits: Focus (14.8%), Creativity (12.9%), Cognition (5.8%) Challenges: Focus (8.8%), Cognitive interference (2.3%)
Anderson et al., 2019	N = 909 Microdosers (Age: 30.5, male: 83%) Non-microdosers (Age: 27.6, male: 9.1%)	LSD (65%) Psilocybin (28%)	Improved: Creativity (more clever, uncommon, & remote)
Cameron et al., 2020	N = 383 (Past or current microdoser) Mean age: 33.26, Male: 64.16%	LSD (48.58%) Psilocybin (26.18%), MDMA (12%), Other (11.51%)	Improved: Memory (38.85%), Attentional focus (58.97%) No Effect: Memory (46.50%), Attentional focus (26.28%) Worsened: Memory (13.10%), Attentional focus (14.74%)
Lea et al., 2020	N = 525 Mean age 34.5 Male: 73.5% Currently microdosing	Psilocybin (46.1%): Up to 0.1g (19.9%), 0.11 - 0.2g (26.8%), 0.21 - 0.35g (22.6%), 0.36 - 0.5g (13.9%), 0.5g+ (12.2%), Unsure (4.5%). LSD (40.38%) (M=13ug) Most common schedule: One day on, two days off (31.8%)	Enhanced cognitive performance: Easier to get "in the zone" (47.8%), Energy (46.5%), Problem-solving (43.2%), Focus (57.7%), Creative (54.1%) Impaired: Concentration (49.9%) Confusion (24.2%)
Petranker et al., 2022	N = 6753 LSD: Median age 23, male: 74% Psilocybin: Median age 26, Male: 71% Both: Median age = 26, Male: 73%	LSD (58%) Psilocybin (29.17%) Both (12.76%)	Benefits: Creativity (LSD: 47%, MM: 49%), Productivity (LSD:36%, MM:34%), Focus (LSD: 33%, MM:30%), Mental clarity (LSD:45%, MM:29%)

			Challenges: Mental confusion (LSD:10%, MM:8%), Focus (LSD 9%, MM:8%)
Study	Sample	Microdosing Practices	Effect on Cognition
Haijen et al., 2022a	ADHD & Intending to MD, Baseline: $N=233$ (116; 35.3 ± 10.7), 2W: $N=64$ (30; 35.4 ± 10.1), 4W: $N=44$ (21; 37.2 ± 9.7)	Psilocybin ($N=91$): 722 mg LSD alternatives: ($N=14$), 17.5 μg LSD ($N=11$): 12 μg	Improved: Attention
Haijen et al., 2023			Improved: Attention & Impulsivity
Haijen et al., 2024	MD: Baseline: $N=180$ (94; 36.3 ± 11), 4W: $N=38$ (16; 38 ± 8.9) ADHD Med: Baseline: $N=37$ (5; 37.4 ± 12.9), 4W: $N=28$ (5; 38.9 ± 13.8)		Decreased: Inattention & hyperactivity/impulsivity (compared to medicated group)

Note. In the Petranker et al. (2022) study, "MM" refers to "magic mushrooms" containing psilocybin. The abbreviation is used to distinguish between reports from participants who used LSD versus those who used psilocybin mushrooms.

Two interview-based studies provide deeper qualitative insights into subjective cognitive experiences, with improvements dominating the reported effects (as detailed in Table 4). Webb et al. (2019) interviewed 30 participants and found consistent reports of enhanced productivity, creativity, energy, and focus. Similarly, Johnstad (2018) interviewed 21 male participants and identified improvements in energy and cognitive function. Additionally, a case report of an individual microdosing ibogaine to help with bipolar disorder documented substantial improvements in mental clarity and organised thinking following microdosing. The interview format provides rich descriptions of cognitive enhancements that aligns with the survey data described above.

Table 4:

Summary of Studies utilising an Interview Design to Investigate the Cognitive Effects of Psychedelic Microdosing

Study	Study design	Sample	Substance	Dose and schedule	Effect on cognition
Fernandes-Nascimento et al., 2022	Case Report	1 female (Aged 47) with bipolar disorder type II	Ibogaine	4mg, Twice a day	Improved: Mental clarity and Organised thinking
Johnstad, 2018	Interview	$N = 21$ males (Median age: 30)	LSD, Psilocybin		Improved: Energy & cognition
Webb et al., 2019	Interview	$N = 30$ (Median age: 31, Male: 66.67%)	Psilocybin ($N=13$), LSD/IP-LSD ($N = 11$), Both ($N = 6$)	Psilocybin: 0.2 & 0.5g LSD: One "drop" of LSD Every three to four days	Improved: Productivity Creativity, Energy, Focus

Qualitative content analyses across multiple studies consistently identified cognitive enhancement as a prominent theme in naturalistic microdosing reports (Table 5 provides an overview of these studies). Analyses of YouTube videos, comments, and Reddit discussions repeatedly found that users highlight improvements in creativity, productivity, focus, problem-solving, and general cognitive function (Andersson & Kjellgren, 2019; Lea et al., 2020; Miura, 2022; Pop et al., 2024). These analyses demonstrate that the microdosing community consistently values and reports cognitive enhancement as a primary benefit, reinforcing the patterns observed in survey and interview studies.

Table 5:

Summary of Studies utilising a Thematic or Content Analysis to Investigate the Cognitive Effects of Psychedelic Microdosing

Study	Study design (Condition)	Sample	Common themes
Andersson & Kjellgren, 2019	Thematic analysis of youtube videos and comments	<i>Videos</i> N= 34 (24 male) <i>Comments</i> N= 198	Improved abilities and optimal performance (commonly increased creativity and productivity)(Increased focus, and motivation, problem solving)
Fadiman & Korb, 2019	Systematic exploration of descriptive reports	Intending to microdose on their own	<i>Change in work:</i> For ADHD/attention issues & to increase productivity and creativity
Lea, Amada, & Jungaberle, 2020	Content analysis of r/microdosing	First 200 Reddit discussion threads	Benefits: Cognitive & creative enhancement Challenges: Limited or no cognitive improvement
Miura, 2022	Quantitative content analysis of r/microdosing	462 Reddit posts	Improved: Creativity, Productivity, Energy
Pop et al., 2024	Thematic analysis of r/microdosing subreddit	all publicly available submissions	Neurocognition, Self enhancement

Collectively, these subjective studies present a striking contrast to laboratory findings. Across all naturalistic research, there is a consistent pattern of cognitive benefits from microdosing. While some challenges like confusion or attention difficulties were reported, they were substantially outnumbered by positive cognitive effects. This disconnect suggests that perceived cognitive benefits may involve elements not captured by standardised

laboratory measures, such as interactions with setting, expectancy, or co-occurring effects in mood and well-being.

Limitations of Objective Research

The psychedelic microdosing literature contains several methodological limitations and flaws. Beginning with the limitations of lab-based studies using objective measures, the duration and number of doses in many studies are not typical of what is seen in the microdosing community. Studies typically only give participants a single dose and then measure aspects of cognition (e.g., Cavanna et al., 2022; Fallon, 2021), whereas, in the microdosing population, it is common for individuals to follow a regime that lasts months or years (Lea, Amada, Jungaberle, et al., 2020a). This discrepancy between acute and long term microdosing represents a fundamental flaw that may explain the contradictory findings between lab-based and naturalistic studies. The cognitive benefits frequently reported by long-term microdosers may emerge gradually through neuroplastic changes that require sustained, repeated exposure rather than occurring immediately after a single dose. Therefore, null findings from lab-based studies should be interpreted cautiously, as focusing exclusively on the acute effects of a single dose could potentially miss cumulative neurobiological changes that underlie cognitive enhancement.

A second limitation in objective research is the universal dose that is administered to participants. This is also not typical of the microdosing population that often displays huge variability in dosing ranges. For example, one naturalistic study reported that 23% of psilocybin users dosed less than 100mg, while 18% reported a dose of more than 500mg (Lea, Amada, Jungaberle, et al., 2020b). The same variability has been identified with LSD doses with some participants taking doses as small as 5ug, while others take doses as high as 32ug (Szigeti et al., 2021). This highlights that the dose consumed varies drastically and

typically is tailored to suit the individual. As a result, objective studies using fixed doses may fail to capture cognitive improvements that only emerge when dosing is tailored to the individual's needs.

A third limitation that may be contributing to the conflicting findings, is the lack of consideration for mental set and environmental setting. Given that set and setting have demonstrated significant importance for the therapeutic effectiveness of macrodoses, it is possible that a laboratory setting may prevent the researchers from observing the noticeable improvement seen in qualitative studies. The relevance of the setting is demonstrated by Murphy and colleagues (2024), where participants completed their first set of cognitive measures in a laboratory, followed by home administration for the following 13 doses. They concluded that ratings of “creativity” and “energy” were smallest on the first day, indicating that participants felt less creative/energetic when given a microdose in a laboratory environment. This suggests that the lab-based environment itself may reduce the effects of microdosing, potentially contributing to the null effects typically reported in objective research

A final limitation of lab-based studies is the predominant use of ‘healthy’ participants. Naturalistic research predominantly consists of participants who report having existing mental health disorders or who have self-identified a deficit in cognition and are microdosing with the intention of improving their symptoms. As seen in the quote “All participants, except Leo, had been wanting to help themselves with an aspect of their lives in particular, whether it be mental health, relationships, cognition, or overall wellbeing.”(Ryan et al., 2023). However, objective studies typically use ‘healthy’ participants who do not have existing mental health conditions and are less likely to present any cognitive deficits. Therefore, these participants likely already have higher baseline cognitive abilities and may not show a

significant improvement because of ceiling effects. For an in-depth critique of the current state of microdosing research, refer to Petranker et al. (2024).

Limitations of Subjective Research

Subjective studies also present many limitations, most notable, is the absence of objective validation. Without standardised cognitive assessments, self-reported improvements lack verification and may be influenced by various confounding factors. Expectancy effects appear particularly influential, as demonstrated by Szigei et al. (2021), who found that creativity improvements were only significant when participants believed they were taking a microdose. Additionally, while offering ecological validity, naturalistic research contains numerous uncontrolled variables that may influence cognitive experiences. For example, many microdosers report changing habits and behaviours to more beneficial practices (e.g., healthier diet, reduced substance use, improved sleep), making it impossible to isolate which benefits are attributable to the microdose itself (Lea, Amada, & Jungaberle, 2020).

Furthermore, self-selection bias is inevitable in naturalistic research, as individuals are not randomly assigned to microdosing, but rather, consists of individuals who already had a desire to microdose. As a result, individuals in naturalistic research are likely approaching the practice with positive expectations and motivations, guiding them to report beneficial outcomes. It should be noted that this bias is also present in the recruitment of lab-based studies where individuals know what they are signing up for, however, null results are systematically reported. On the contrary, in real world contexts individuals who experience no effects from microdosing are under no obligation to discuss them before leaving the practice, leading to more extreme outcomes in naturalistic research.

Finally, another limitation in subjective studies is the variability of potency in each microdose. Studies have concluded that the concentration of psilocybin in the same fruiting

body can vary up to 400% (Beug & Bigwood, 1982). Additionally, the way a substance is stored has an effect on potency such that exposure to light and heat can substantially reduce psilocin (the active metabolite in psilocybin) potency over time, thus leading to unidentifiable doses (Gotvaldová et al., 2021). These uncertainties make it difficult to control for microdosing dosage in subjective research.

Overall, a limitation that applies to both objective and subjective studies in the existing literature is that cognitive effects are typically not the main focus of investigation, but is rather only measured as a secondary outcome or supplementary analysis. Therefore, the assessment of cognitive changes is often less comprehensive, with studies using few cognitive measures or failing to assess the full range of cognitive domains. Future research specifically designed to examine cognitive effects as the primary outcome would likely provide more definitive evidence regarding the impact of microdosing on various domains of cognitive functioning.

Future Directions for Microdosing Research

Despite the limitations in current microdosing research, both objective and subjective approaches are necessary to offer a comprehensive view of the effects of microdosing. Lab-based studies offer control and objective measurement necessary for isolating the effects of microdosing. While naturalistic studies compliment and guide this research by providing ecological validity, and insights into how cognitive effects manifest in everyday life.

Moving forward, research should address the disconnect between objective performance and subjective experience, as this discrepancy likely reflects more than methodological differences. While laboratory studies provide essential controlled assessment, they may fail to identify how cognitive effects manifest in real-world contexts. Most notably, controlled studies treat microdosing as an isolated practice, however, real-world microdosing

rarely occurs in isolation. Instead, microdosing typically occurs in combination with multiple other lifestyle changes such as meditation, increased socialising, journaling, and other wellness behaviours (Andersson & Kjellgren, 2019). These additional practices may work in combination with microdosing to produce cognitive changes, therefore, neglecting them in controlled studies would be failing to capture the whole practice. Future research should therefore investigate not just whether microdosing affects cognition, but how it interacts with these co-occurring practices to potentially influence cognitive changes. Understanding this complex dynamic is essential for bridging the gap between laboratory findings and lived experience.

The Present Study

Our study addresses the disparity in existing research by employing a mixed-methods design to investigate the effects of psychedelic microdosing on cognitive functioning and potential mediating factors. We selected the social media platform Reddit as our primary data source, which addresses challenges associated with lab-based studies. Reddit offers easy access to hard-to-reach participants, the ability to investigate phenomena in a natural setting, and savings in time and costs (Rocha-Silva et al., 2024). In addition, unlike other social media platforms, Reddit provides greater anonymity as users are not required to include identifying information on their profiles. The ability to remain anonymous online is suggested to increase the likelihood of speaking freely on social media (Amichai-Hamburger, 2013). This makes it a particularly valuable platform for research on sensitive topics such as microdosing, as psychedelics are classified as controlled or prohibited substances in most countries worldwide (UNODC, 2013). The effectiveness of social media data in psychopharmacology research has been demonstrated in drug safety studies, where black box warnings, labelling changes, or withdrawals could be detected three months to nine years in advance (Lee et al., 2021).

Our study employed both passive and active data collection approaches. For our primary data source, we utilised passive data collection by extracting posts from the r/microdosing subreddit. We selected this subreddit due to its substantial user base and focus on microdosing practices and outcomes, providing an ideal source of naturalistic data. This passive approach extracted content that was created by "unintentional participants" who were not specifically recruited for research purposes (Rocha-Silva et al., 2024), capturing authentic discussions about cognitive effects in a context without researcher intervention. In addition, this approach allowed for quick data collection as the posts were readily available. We extracted posts mentioning cognitive changes using systematic filtering methods and analysed them through quantitative content analysis.

To complement this passive approach, we also employed an active data collection strategy through our exploratory survey. The active data collection approach involved direct contact with participants and informed them that their responses would be used for research purposes (Schwab-Reese et al., 2018). We recruited participants by posting survey links on various microdosing related subreddits, allowing interested users to voluntarily participate. The survey enabled us to ask targeted questions about specific cognitive domains, contextual factors, and other subjective experiences that might be underreported in spontaneous Reddit posts.

This dual approach offers several advantages. The passively collected Reddit data captures authentic, unprompted experiences, while the actively collected survey data allows for more structured investigation of cognitive effects and potential mediating factors. By combining these methods, we create a more comprehensive understanding of how microdosing affects cognitive functioning in real world settings.

Previous microdosing studies have used Reddit to recruit participants for interviews and collect data for textual analyses (Lea, Amada, & Jungaberle, 2020; Miura, 2022; Pop et al., 2024a; Ryan et al., 2023). Research by Ryan et al. (2023), Pop et al. (2024), and Lea et al. (2020) identified common themes in microdosing discussions, regarding users motivations, practices, and outcomes. While these studies have identified themes related to microdosing and cognition, they have not specifically focused on the cognitive effects and their mediating factors. Our study addresses this gap by specifically examining how microdosing affects cognitive functioning and what factors might mediate these effects. Upon identifying key themes associated with cognitive outcomes from both data sources, we explore how these factors may influence cognitive changes. This approach provides ecological validity that laboratory studies often lack, while also allowing us to research psychedelics in a country where these substances remain criminalised.

To summarise, our study combines passive and active data collection methods to investigate the effects of psychedelic microdosing on cognitive functioning and potential mediating factors. To investigate this relationship, we perform a qualitative and quantitative analysis. Together, these complementary approaches offer unique insights into the complex relationship between microdosing and cognitive functioning.

Chapter Two: Method

Our study primarily analysed naturalistic Reddit data to examine the cognitive effects of microdosing. This was supplemented by an exploratory survey, designed to probe deeper into the patterns identified in the Reddit analysis. The Reddit data captures authentic and unprompted discussions of individuals' experiences with microdosing. Our exploratory survey complemented the Reddit analysis by allowing us to ask directed questions about specific cognitive effects, contextual factors, and other subjective domains associated with microdosing that are often underreported in spontaneous Reddit posts. This structured approach enabled us to examine potential relationships between cognitive effects and other qualitative changes, as Reddit users typically do not report all aspects of their microdosing experience. For example, while a Reddit user might mention a change in cognition (e.g., increased ability to focus), they may not discuss co-occurring changes in other domains that could be mediating these effects (e.g., reduced substance use). Additionally, given the limited research on the relationship between cognition and microdosing, the survey aimed to help establish the foundation for future standardised measures. The following sections detail our methodology for both the Reddit analysis and survey components of the study.

Part 1: Reddit Data

Data Source

Data for this study was retrieved from the subreddit “Microdosing: sub-threshold dosing of psychedelic drugs for self-improvement, therapy or well-being” (r/microdosing). This is a popular subreddit that invites discussion on microdosing research, experiments, regimens and experiences. The subreddit was founded on October 16, 2013, and by September 16, 2024, it counted 275,499 members (*r/microdosing*, 2024).

Data Collection and Preprocessing

Following the methodological approach established by Wightman and colleagues (2023), our initial step for obtaining data relevant to microdosing involved extracting data from the entirety of the r/microdosing subreddit. The r/microdosing subreddit was selected as our primary data source due to its abundance of content specifically on microdosing practices and outcomes. We obtained all retrievable data from the r/microdosing subreddit from October 2013 to December 2024 via Academic Torrents ($N = 57,902$), which includes posts, comments, and their meta data (Cohen and Lo 2014; Lo and Cohen 2016). We used Academic Torrents for data collection in our study due to the recent restrictions implemented to PRAW (Reddit's official API) in June 2023, which limited access to the most recent 1,000 posts (Reddit, 2023). This data repository enabled us to analyse a much larger and historically complete dataset than would otherwise be possible with current API limitations. Our systematic approach to data collection is illustrated in Figure 1.

After extracting posts from Academic Torrents, we pre-processed the data via multiple steps. Initially, we removed Reddit formatting elements and special characters from the raw text data. We then applied our inclusion criteria requiring posts to: (1) contain textual content beyond the title (e.g., excluding posts with images only), (2) be written in English, (3) explicitly reference a psychedelic substance, (4) mention microdosing, and (5) discuss an effect on cognitive functioning. Posts from moderators and those with deleted or removed content were also excluded.

Due to the extensiveness of our data set, we developed custom python scripts to efficiently apply our inclusion/exclusion criteria. To identify posts containing psychedelic substances, one script extracted posts using a comprehensive list of psychedelic substances. A separate script identified microdosing related posts by detecting explicit terminology (e.g.,

“microdose”, “md”, “sub-perceptual”), and dosage patterns. This dosage identification process extracted posts mentioning microdosing ranges while excluding those describing macrodosing ranges, with thresholds based on prior literature (Johnstad, 2018). The custom python scripts used are available upon request.

To identify posts discussing an effect on cognitive functioning, we implemented a two-stage natural language processing approach. In the first stage we filtered for cognition related posts by searching for keywords in the title and text of each post. These keywords were derived from three sources: (1) findings from previous microdosing literature, (2) established cognitive assessment scales, and (3) exploratory coding of a pilot sample from our data set. Combining these three sources ensured our keyword list captures both standardised cognitive terms and unique language patterns found in Reddit discussions.

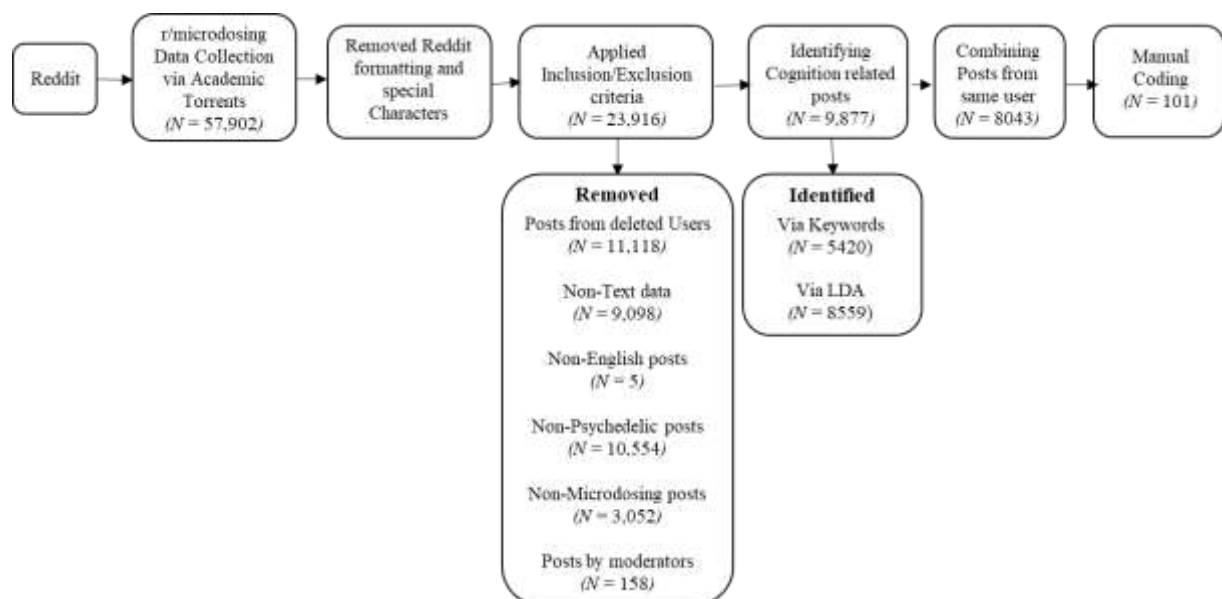
The second stage employed Latent Dirichlet Allocation (LDA), a topic modelling technique that identifies patterns of word co-occurrence to discover underlying themes in text data. This approach helped us capture cognition-related posts that did not contain our predefined keywords but still discuss cognitive effects. To prepare the data for LDA analysis, text preprocessing included lowercasing, removal of punctuation and whitespace, tokenisation, stopword removal, and lemmatisation. We created a document term matrix, filtering to retain words that appeared in at least 25 posts (At least 1% of posts) but no more than 95% of posts to reduce noise. The LDA model was trained to identify 100 topics using symmetric alpha priors and was run 25 times over the complete dataset (passes = 25) to ensure reliable topic identification. We tested different numbers of passes and selected 25 as the optimal number based on the clarity and interpretability of the resulting cognition related topics.

Initial topics were generated through LDA analysis, after which we manually reviewed each theme and its associated keywords to assess its relevance to cognitive functioning. Based on this review, we developed a topic weighting system with three tiers: primary cognitive topics (e.g., problem solving and concentration) received the highest weight (0.4), broader cognitive-related topics (e.g., confusion) received a moderate weight (0.3), and peripherally related topics (e.g., academic performance) received the lowest weight (0.2). This weighting system allowed us to quantify how strongly each post related to cognitive functioning. We then calculated a cognitive relevance score for each post by combining two measures: the frequency of cognitive-related keywords in the text and the post's association with cognitive related topics. Posts were included in our final analysis if their relevance score fell within the top 25% of all scores in our dataset.

To prevent individual users from disproportionately influencing our results, we combined multiple posts from the same user. The progressive reduction in post count after each processing step is documented below in Figure 1.

Figure 1:

Flowchart of Data Extraction and Processing from r/microdosing Subreddit



Procedure

To investigate how microdosing psychedelics affect cognitive functions and identify potential mediating factors, a qualitative content analysis of the r/microdosing subreddit was performed using NVivo14 (version 14.23.3(46)). A systematic coding approach was followed when analysing the data, such that variables of interest were determined prior to coding. To refine the coding variables and ensure the validity of the codebook, three coders independently coded 15 Reddit posts according to the initial codebook. The final codebook used for data analysis is displayed in Table 6.

To select posts for manual analysis, we utilised the engagement-based ranking system described in Wightman and colleagues (2023), where posts were ranked according to their 'net score,' which represents the difference between a posts upvotes and downvotes. Following this score-based ranking, we accumulated the top 300 scoring posts and read through them in their entirety to ensure they met the inclusion criterion. Of these 300 posts, only 101 met the inclusion criteria. The posts that did not meet the criteria were commonly excluded because they did not mention a change in cognition, or the post did not report a microdosing experience (e.g., inquiry or informational posts). The 101 posts were manually coded according to the predefined codebook using NVivo14. In cases where coding classifications were ambiguous, the sentences were flagged for review and subsequently validated through consultation with the research supervisors, thus enhancing coding reliability.

The 8400 posts that were not included in the manual coding were to be automatically coded using supervised machine learning techniques as described by Wightman and colleagues (2023). This would have allowed for automatic classification based on the patterns discovered in the manually coded data. However, despite multiple attempts with different

types of machine learning, the outputs were not able to demonstrate adequate consistency with the manually coded data. As a result, our final analysis relies exclusively on the 101 posts that were coded manually, as we were unable to produce valid codes for the remaining dataset.

Measures

Reddit Codebook.

Our codebook (Table 6) was organised hierarchically using a structured sub-coding framework to allow for systematic analysis. Following established qualitative research practices (Gibbs, 2018; Miles et al., 2020), we used a multi-level coding hierarchy with broad 'parent' codes representing overall themes, and multiple levels of children codes to enrich each data entry. Specifically, our codebook utilised four levels: themes, subthemes, levels, and codes. This hierarchical organisation follows what Saldaña (2021) describes as 'splitting' rather than 'lumping' of data, allowing us to conduct detailed analysis at granular levels while still maintaining the ability to examine broader thematic patterns across the dataset.

The development of the codebook was progressively refined, consisting of two phases that combined a deductive and an inductive approach. During the first phase (deductive), initial themes and subthemes were developed based on the existing microdosing literature. In the second phase (inductive), themes and subthemes were refined by conducting an exploratory coding phase with a small sample of Reddit posts. The refinement process consisted of deleting, merging, renaming, and expanding codes. This approach to codebook development has been used in existing literature and enables methodological flexibility (see Shaw et al., 2024). As an additional measure, we observed initial survey responses for codebook validation. Several subcodes (dosage, duration, additional substances/stacking, age, and diagnosis) were initially open coded rather than being predefined. This preserved the

details of the original data which were then categorised into meaningful bins based on their distribution patterns.

Table 6:

Psychedelic Microdosing Codebook

Theme	Sub-Theme	Description	Levels	Example codes
Affective Changes	Mood States:	Changes in emotional states and mood experienced during or after microdosing.	•Improved •No Effect •Worsened	•Happiness •Sadness •Relaxed •Anger
	Emotion Regulation:	Changes in ability to manage, process, and respond to emotions during or after microdosing.	•Improved •Worsened	•Emotional Control •Emotional Resilience •Emotional Awareness
Cognitive Changes	Higher Order Cognition:	Changes in complex thinking processes and advanced cognitive functions	•Improved •No Effect •Worsened	•Abstract Thinking •Problem Solving. •Creative Thinking •Decision Making
	Attentional Processes:	Changes in ability to focus, concentrate, and maintain attention on tasks or experiences.	•Improved •No Effect •Worsened	•Focus •Task Switching
	Memory and Learning	Changes in ability to retain information, learn new concepts, or recall existing knowledge.	•Improved •Worsened	•Short-term Memory •Learning Speed. •Long-term Memory •Information Retention
	Cognitive Clarity:	Changes in mental clarity and thought organisation processes.	•Improved •Worsened	•Thought Organisation •Mental Clarity
	Language Processing	Changes in cognitive processes related to language production and comprehension	•Improved •Worsened	•Articulation •Comprehension
	General Cognition	Broad changes in cognitive functioning. This category captures non-specific cognitive changes that does not fit a specific cognitive domain.	•Improved •No Effect •Worsened	•Overall mental performance •Mental Speed •Productivity •Confused
Demographics	Age	What is the age of the participant.	Numerical variable	•Record age when given
	Gender	What is the participants gender identity.	•Male •Female •Non-binary	
	Diagnosis	What medical or psychological diagnoses does the participant report having?	Open ended qualitative •Record diagnosis when given	•Autoimmune disease •Psychiatric Disorder •Neurological Disorder •Physiological Disorder
	Occupation	What is the Individuals occupation?	Open ended qualitative	•Record occupation when given
	Environment	environmental factors that may influence the participant's microdosing experience.	•Negative •Positive	
	Family	Participant's relationship status and family structure.	•Marital status •Children	
	Current habits	What habits and behaviours does the participant currently engage in.	•Positive habits •Negative habits	
	Prior to microdosing	Participant's experiences or treatments prior to microdosing that may influence outcomes.		•Has macrodosed •Medication/Therpay Didn't Work •Negative History
Effect on Health Conditions	Change in Health outcome	Changes in symptoms or management of specific health conditions reported during or after microdosing.	•Improved •No Effect •Worsened	
	Disease insight:	Changes in understanding or perception of one's health conditions resulting from microdosing.	•Improved •Worsened	•Symptom Awareness •Management Strategy •Condition Impact
Insights and Transformation	Self-Awareness and Personal Growth:	Changes in self-understanding and personal development attributed to microdosing.	•Improved •Worsened	•Self-Image •Self-Understanding •Personal growth
	Lifestyle and Habit Changes:	Changes in behaviours and daily practices resulting from microdosing experiences.	•Positive changes •Negative changes •Neutral changes	•Reduced Bad Habits •New/increased Habits •Neutral changes
	Spiritual and Existential Experiences:	Changes in experiences related to spirituality.	•More meaningful •Less meaningful	•Spiritual Connection •Nature connection. •Life Purpose •Meditation
	Perspective Shifts:	Changes in how participants view life and their place in it following microdosing experiences.	•Positive shifts •Negative shifts	•Life Attitude. •Future Outlook. •Values Alignment •Life Appreciation
	Psychological Processing:	Changes in processing of psychological material and experiences attributed to microdosing.	•Improved •Worsened	•Pattern Recognition •Emotional Understanding •Coping Development •Personal History Integration
	Mindfulness and Present Moment Awareness:	Changes in present-moment awareness and mindfulness capabilities during or after microdosing.	•Increased •Decreased	•Sensory/Environmental Awareness •Present Focus
Microdosing Approaches	Dosage	What quantity of the substance does the person ingest each microdose.	Numerical	•Report in micrograms (mcg) for LSD and milligrams (mg) for psilocybin.
	Regime/Schedule/ Protocol	How frequently does the individual microdose.	Open ended Qualitative	•Record regime when given
	Substance	What substance is used for microdosing.	•LSD •1P-LSD •Psilocybin •Mescaline	•Report species of mushrooms if stated
	Duration	For how long has the individual been engaging in this microdosing practice.	Numerical	•Record duration when given

Theme	Sub-Theme	Description	Levels	Example codes
	Additional Substances/Stacking	What other substances does the individual report ingesting on the day(s) they microdose	Open ended Qualitative	• Record substance when given
	Effects Felt	Does the individual report intoxicating effects	•Yes •No	
	Microdosing Adjustments	Changes to dosage or regimen made by the individual.	•Changed Regime •Changed Dose	•Mentioned with effects (Describes the effects of two different doses) •Mentioned without effects (Describes the effects of one dose)
	Pre Conception	Participant's attitudes toward microdosing before starting the practice.		•Sceptical •Hopeful. •Fearful •Curious •Reluctant
	Setting	What context and environment does the microdosing occur.	• Work • Home life • Studying • Travelling • Social • Sports/recreation	
	Time of Dose	What time of day does the individual typically take their microdose.	•Morning •Evening	
	Stopped microdosing	Information about cessation of microdosing and any lingering effects reported.	•Took a break (effects Mentioned) •Stopped (effects mentioned)	•Effects remain positive. •No effects mentioned. •Some effects remain. •Went back to baseline
Motives/ Intentions	Treatment for Medical condition:	Specifically mentions microdosing to address psychological or medical condition.		
	General Wellbeing improvement:	Specifically mentions microdosing for self enhancement and improved quality of life.		
	Cognitive Enhancement:	Specifically mentions microdosing to improve mental performance.		
	Curiosity:	Specifically mentions microdosing out of curiosity without specific therapeutic or enhancement goals.		
	Spirituality:	Specifically mentions microdosing for spiritual or mystical development		
	Physical Health Improvement :	Specifically mentions microdosing to improve physical health and bodily wellness.		
	Substance Use Reduction:	Specifically mentions microdosing to reduce use of other substances.		
	Social Enhancement:	Specifically mentions microdosing to improve social relationships.		
Other	Brain changes	Specifically mentions microdosing to improve neurological structure or function.		
	External Observation	Changes in the individual noticed and reported by friends, family members, colleagues or other external observers.	•Negative Change •Positive Change	
	Information Source	Where did the individual find information about microdosing	•Scientific literature •Online Communities •Personal Networks •Media Content •Informed, unspecified. •Uninformed	
	Sentiment Rating	What is the overall emotional tone of the post	1 = Really Negative 2 = Negative 3 = Neutral 4 = Positive 5 = Really Positive	
	Information Rating	How useful and informative is this post	1 = Really Uninformative 2 = Uninformative 3 = Neutral 4 = informative 5 = Really Informative	
Physiological Changes	Energy:	Changes in physical and mental energy levels during or after microdosing.	•Improved •Worsened	
	Sleep	Changes in sleep patterns and quality attributed to microdosing.	•Improved •Worsened •Dream related	
	Sensory and Perceptual changes:	Changes in how senses (sight, sound, touch, smell, taste) are experienced during or after microdosing.	•Improved •Worsened	•Sight •Sound •Touch •Smell •Taste
	Physical Performance:	Changes in physical capabilities and performance resulting from microdosing.	•Improved •Worsened	
	Physiological Discomfort:	Changes related to physical discomfort or unusual bodily sensations during or after microdosing.	•Improved •Worsened	•Gastrointestinal related •Head-Related •Other
Prior Cognition		How was the individuals baseline cognitive functioning before starting microdosing.	•Impaired •Enhanced	•Higher Order Cognition •Cognitive Clarity •Attentional Processes •General Cognition •Memory and Learning

Theme	Sub-Theme	Description	Levels		Example codes
Social Changes	Social Engagement:	Changes in patterns of social interaction attributed to microdosing.	•Increased	•Decreased	
	Interpersonal Skills	Changes in ability to interact and communicate effectively with others in social contexts during or after microdosing.	•Improved	•Worsened	
	Emotional Connection	Changes in ability to form and maintain emotional bonds with others, including empathy, attributed to microdosing.	•Improved	•Worsened	
	Social Perception	Changes in how social situations are perceived and interpreted during or after microdosing.	•Improved	•Worsened	
Temporal Effect	Day of Effect	What day (dosing or non-dosing) were the effects experienced?			•Dosing Day •Non dosing day
	Mixed Effects	Did the effects change over time			•Different effect (Same day) •Different effect (different days) •Different effects over time (Tolerance)

Part 2: Survey Data

Sample and Recruitment

Primary recruitment for our survey occurred through posts on the online forum ‘Reddit’ (Reddit Inc, San Francisco, CA, USA). Links were posted under the username u/Microdosing_Research on the following psychedelic-related subreddits: r/LSD, r/microdosing, r/Psychedelic, r/Psychedelics, r/Psychonaut, r/RationalPsychonaut, r/Shrooms. Approval was requested to subreddit moderators prior to posting to respect subreddits rules regarding research recruitment and to ensure transparency with the moderators regarding research objectives. Additional participants were recruited by advertising the study in a section of the Student Association for Psychedelic Investigation (SAPI) club newsletter at Victoria University of Wellington. In addition, we also recruited participants through direct outreach to known microdosers. Only participants with experience in microdosing psychedelics and English fluency were eligible to participate in this study. Participation in the survey was anonymous and voluntary. A total of 29 responses were received.

Procedure

The survey was created using Qualtrics and was made available from December 5, 2024, remaining active for four months. The recruitment posts on each subreddit provided comprehensive information about the study, including the research team, the study’s purpose, and eligibility criteria (Recruitment post and information sheet can be seen in Appendix A-

B). Individuals who clicked the survey link were directed to an information sheet that detailed the study's background, objectives, and ethical considerations. Participants were informed about the survey's content (demographics, microdosing practices, and experienced effects), the voluntary nature of participation, and the anonymity of responses provided. The information sheet also displayed contact details for addressing any questions about the research. Informed consent was obtained electronically prior to participation, with users indicating their agreement to participate by clicking the forward arrow at the bottom of the information sheet. The survey was structured into eight blocks to reduce fatigue and maximise quality. To enhance participant engagement, visual elements were presented prior to each block, and a progress bar was displayed throughout the survey. While questions were presented in a fixed order to ensure logical flow, participants were able to skip questions and return to previous ones, supporting voluntary participation.

Measures

Survey Questions.

The survey was split into eight blocks, each capturing a different topic related to microdosing. Block one (Demographics and Background) consisted of eight demographic questions to identify microdosing history, age, gender, country of origin, daily occupation, level of education, height, and weight (used to calculate BMI). Block two (Health History) contained four questions that asked participants about any health diagnoses they had received, what age they received the diagnosis, and how microdosing has affected their symptoms. Block three (Substance Use Patterns) contained two questions that asked participants which substances they had used while microdosing, and how frequently they had used each specified substance. Block four (Microdosing Practice) contained five questions focusing on the duration of microdosing, whether they had taken any breaks, motivation to start microdosing, how they microdose (e.g., substance, dose, frequency), and how they

decided on each component of their microdose. Block five (Effects Timeline) contained one question asking participants to describe the pattern of microdosing effects (e.g., how the effects change throughout the day and how long they last). Block six (Cognitive Effects) contained two questions that asked participants about their cognitive abilities in certain cognitive domains (e.g., focus and memory) before and after they began microdosing. Block seven (Broader Effects) contained five questions that aimed to capture whether participants experienced any changes in emotional, physical, social, and lifestyle domains. Block eight (Challenges and Final Thoughts) contained four questions asking whether participants had experienced any unwanted effects from microdosing, what these effects were, and how they think microdosing contributed. Participants were also given an opportunity to mention anything else about their microdosing experience they thought was important.

The topics of each block were derived from the codebook and preliminary analysis of Reddit data, representing a deductive approach to question development. The questions were specifically designed to serve as a foundation for item development, which is the first phase in scale development and validation (Boateng et al., 2018). Qualitative data can be used to inductively identify domain items (see Boateng et al., 2018), therefore, questions were intentionally kept open-ended, allowing participants to describe their experiences in their own words without being constrained by predetermined response options. While the primary purpose of open-ended responses is to supplement the findings from the Reddit data, participants' responses could potentially be utilised by future researchers to identify key themes and develop specific items and subscales for standardised instruments. This inductive approach would ensure that future quantitative measures are grounded in the lived experiences of microdosers rather than solely theoretical frameworks.

Data Analysis

Data analysis followed a systematic quantitative content analysis approach. For the Reddit data, after coding was completed using NVivo14, descriptive statistics (frequency counts and percentages) were generated for each variable in our codebook. Results are presented as frequency distributions and descriptive statistics, highlighting the most commonly reported cognitive effects and their contextual characteristics. The material within each code was reviewed to identify representative quotes. Selected quotes were paraphrased, which is in line with ethical guidance and recent conventions when using Reddit data, to reduce the risk of reverse identification (Proferes et al., 2021).

For the survey data, open ended responses were systematically reviewed to identify prominent patterns. These patterns were categorised using the same codebook structure developed for the Reddit analysis, allowing for direct comparison between data sources. Descriptive statistics (frequency counts and percentages) were calculated for each identified theme. Typographical errors were corrected in all reported quotations from survey responses.

Ethical Considerations

This study utilised publicly available data from Reddit. To protect user privacy, all identifying information was removed during the data cleaning process. Ethics was approved for our study by the Human ethics Committee at Victoria University of Wellington (Approval number: 31472).

Chapter Three: Results

Part 1: Reddit Results

Content analysis of 101 Reddit posts highlighted several key themes related to users' microdosing experiences. To identify meaningful patterns, we selected themes that were present in at least one-third of the total posts (≥ 33 posts). This threshold helps identify experiences shared frequently enough to suggest potential associations with cognitive changes, while excluding uncommon or isolated effects. The following results section examines these high-prevalence themes that warrant deeper investigation into their potential relationship with cognitive changes. Key themes and their frequencies are presented in Table 7, which summarises the most prevalent findings from our content analysis. The comprehensive quantitative content analysis with all identified themes and their respective frequencies is available in Appendix C.

Table 7:

Reddit Content Analysis: Key Themes and Frequencies

Theme	%	Sub-Theme	%	Level	%	Code	%
Affective Changes	74%	Mood States	67%	Improved	61%	Happiness	42%
		Emotion Regulation	35%	Improved	30%		
Cognitive Changes	100%	Attentional Processes	50%	Improved	39%	Focus	38%
		Cognitive Clarity	41%	Improved	35%	Mental Clarity	27%
		Higher Order Cognition	33%	Improved	29%		
Demographics	85%	Health Diagnosis	55%	Psychiatric Disorder	48%	Depression	34%
		Prior to microdosing	56%			Negative history	36%
Insights and transformations	85%	Lifestyle and habit changes	47%	Positive changes	42%	New/Increase positive habits	28%
						Stop/Decrease negative habits	28%
		Perspective shift	42%	Positive shift	41%	Life appreciation	22%
						Life Attitude	23%

Theme	%	Sub-Theme	%	Level	%	Code	%
Microdosing approaches	100%	Self Awareness & Personal Growth	58%	Improved	53%	Personal growth	33%
		Duration	72%	Less than 1 month	34%		
		Regime	55%	3-4 times per week	34%	Every third day	17%
Physiological changes	56%	Substance	86%	Psilocybin	61%		
Social Changes	42%	Energy	29%				
Overall Ratings		Emotional Connection	22%				
		Informational ratings	100%			1	25%
						2	26%
						3	26%
		Sentiment Rating	100%			4	64%
						5	21%

Note. This table presents themes mentioned in at least one-third of posts (>33%), with a few exceptions where key subthemes are included for context. Percentages represent the proportion of the total 101 Reddit posts.

Demographics: Psychiatric Disorders and Negative Life Experiences Frequently Reported

Demographics were commonly reported across the Reddit dataset. The most frequently mentioned demographic subthemes were health diagnoses and attributes prior to microdosing. Among health diagnoses, psychiatric disorders were the most prominent, with depression being the most frequently mentioned condition. For attributes prior to microdosing, individuals frequently reported negative life experiences prior to engaging in their microdosing practice (e.g., “My job performance was terrible and I hated going to work. I stopped exercising and seeing friends and family. My house was a mess and I couldn’t bring myself to cook, so I just ordered takeout constantly. I accumulated significant debt from impulsive purchases of unnecessary things”). These findings suggest that many individuals may turn to microdosing following periods of psychological distress or negative life circumstances.

Cognitive Changes: Improved Attention and Clarity Dominate Cognitive Effects

All posts mentioned a change in cognition, consistent with our inclusion criteria. The most frequently reported cognitive subthemes were changes in attentional processes, cognitive clarity, and higher order cognition. Within attentional processes, the most frequently mentioned change was an improvement in focus (e.g., “I can focus for hours on things I never could before”). Posts that mentioned a change in cognitive clarity, typically report an improvement (e.g., “I noticed my mind getting quieter and realised this isn’t normal for me. Usually I have several thought layers running in parallel, but now there’s just one main channel”). Similarly, posts that mention a change in higher order cognition typically report an improvement (e.g., “Problems have become easier to solve because my imagination is more powerful”), however, the total number of posts reporting this change did not meet the 33% criteria. This trend toward improved higher order cognition is noteworthy due to the large proportion of individuals within the subgroup who reported a positive change (87.9%).

Affective changes: Enhanced Mood and Emotional Regulation Reported

Affective changes were reported in the majority of posts, with both mood states and emotion regulation emerging as prominent subthemes. Changes in mood states were frequently mentioned, with most users reporting improvements, particularly increased happiness (e.g., “I’ve never been happier. I rarely have negative thoughts now”). Changes in emotion regulation were also discussed, with most users reporting an improvement (e.g., “In situations where I would have snapped in anger, I’m now calm and thinking positively”). However, the total number of posts reporting improved emotion regulation did not meet the 33% criteria. This trend toward improved emotion regulation is noteworthy due to the large proportion of individuals within the subgroup who reported a positive change (85.7%).

Microdosing approaches: Psilocybin, Variable Dosing, and Short Durations

All but one post contained information on microdosing approaches. The most frequently mentioned aspects of microdosing were the substance used, dosage, regime followed, and the duration of microdosing. Psilocybin was the only substance mentioned in a substantial number of posts, with most users also specifying their dosage amount. Among those that mentioned psilocybin dosage, the most frequently reported dose was between 0 and 200mg ($M = 178.74\text{mg}$, $SD = 125.13$). Regarding regimen, users frequently reported following a protocol involving three to four doses per week, with the most commonly reporting dosing protocol being every third day (did not meet the 33% criteria). Finally, most users reported relatively short microdosing durations with the majority of users indicating a duration of less than one month.

Insights and Transformation: Growth, Perspective Shift and Lifestyle Improvements

A substantial number of posts mentioned an insight and/or transformation while microdosing. The most frequently mentioned subthemes were changes in lifestyle and/or habits, self-awareness and personal growth, and a shift in perspective. Many users reported a positive change in their lifestyle or habits, which consisted of both starting/increasing positive habits (e.g., “I exercise much more now. From about two days weekly before microdosing to four days weekly after”), as well as stopping/reducing negative habits (e.g., “I quit my social media addiction, stopped smoking cigarettes, and stopped using cannabis entirely”), however neither code met the 33% criteria. Many users mentioned a positive shift in perspective since microdosing, which predominantly consisted of an improved attitude and appreciation towards life (e.g., “I used to count hours until bedtime to escape reality. Now days feel too short. I feel better than I have in years. Microdosing improved my life tenfold”), however these codes did not meet the 33% criteria. Users frequently reported an improvement in self-awareness and personal growth, with personal growth being the most

frequently mentioned improvement (e.g., “I’m more confident than I have been in years”).

Overall Ratings: Positive Experiences Despite Varying Information Quality

Every post was evaluated and labelled with an informational rating and a sentiment rating. The informational rating assessed how well the post described the effects and mechanisms of microdosing, while the sentiment rating measured how negatively or positively the redditor viewed their microdosing experience. Both ratings used a 1-5 scale, where one represented minimal information or a really negative experience, and five indicated comprehensive information or a very positive experience. Regarding sentiment ratings, the majority of posts were scored a four or five, indicating positive attitudes towards microdosing. Regarding information ratings, there was a trend towards posts being rated as uninformative with the majority being scored between one and three. Table 8 reports representative comments for each sentiment rating.

Table 8:

Representative Comments for Each Level of the Sentiment Rating Scale

Sentiment score	Representative comment
1	“No improvements and I’m actually feeling worse”
2	“it quiets my thoughts temporarily like a bandaid but doesn't seem to help me in the long term”
3	“I often forget doses now but it doesn’t matter. I don’t feel different whether I take it or not”
4	“The short version is: it helped a lot but didn't turn my life around”
5	“I’m sharing my story to help others curious about trying it. It works! Psilocybin saved my life despite my initial scepticism”

Part 2: Survey Results

Our exploratory survey gathered data from 29 Participants, although one respondent was excluded from analyses as they referred to microdosing cannabis rather than a psychedelic.

Due to varying completion rates across questions (with only 11 participants completing the entire survey), our analysis includes responses from all participants, and includes proportions relative to the response count for each item. Participants who skipped questions or misinterpreted them were removed from the respective analyses, with frequencies calculated relative to the number of valid responses. Figure 2 illustrates participant attrition throughout the survey. Consistent with our approach to the Reddit data, we focus on themes appearing in at least one-third of responses to identify potentially meaningful patterns related to cognition. Key themes and their frequencies are presented in Table 9, which summarises the most prevalent responses from our exploratory survey. The complete dataset of survey responses is available upon request.

Figure 2:

Participant Attrition Throughout Survey



Note. The percentages displayed above the black line represent the survey completion progress bar. Each blue box indicates participant dropout at different points in the survey, with the number on top showing how many participants left at that point and the number in brackets showing the remaining total. The survey began with 28 participants and ended with 11 completions.

Table 9:*Survey Analysis: Key Themes and Frequencies*

Theme	Sub-Theme	Level	%	Code	%
Affective Changes		Improved	82%		
Cognitive Changes	Focus	Improved	79%		
	Memory & learning	Improved	57%		
		No effect	36%		
	Problem solving & creativity	Improved	73%		
	Clarity of thought & clear-headedness	Improved	80%		
	Organisation skills	Improved	46%		
		No effect	46%		
Prior Cognition	Focus	Poor	87%		
	Memory & learning	Good	40%		
	Problem solving & creativity	Good	60%		
	Clarity of thought & clear-headedness	Poor	57%		
	Organisation skills	Poor	67%		
Demographics	Gender	Male	79%		
	Country	USA	33%		
	Diagnosis	No Diagnosis	33%		
		Psychiatric Disorder	58%	Depression	38%
				ADHD	42%
				Anxiety	33%
Effect on Health Condition	Depression	Improved	100%		
	ADHD	No effect	57%		
	Anxiety	Improved	83%		
Insights & transformations	Lifestyle & habit changes	Improved	89%		
	Personal growth/perspective shift	Positive Change	82%		
Microdosing approaches	Duration	1-6 months	42%		
		>6 months	47%		
	Regime	3-4 times a week	47%	Every third day	26%
	Substance	Psilocybin	68%		
		LSD	32%		
	Additional substances	Caffeine	86%		
		Cannabis	62%		
		Alcohol	48%		
		Psilocybin	38%		
		Nicotine	33%		
		LSD	33%		
		Medication	43%	ADHD Medication	29%
	Additional practices	Exercise	58%		
		Mediation	33%		
Physiological changes	Energy	Improved	73%		
	Sleep	Improved	45%		
	Sensory	Improved	64%		

Theme	Sub-Theme	Level	%	Code	%
Physiological changes	Athletic & physical performance	No effect	36%		
		No effect	82%		
Social Changes	Physical sensations	No effect	55%		
		Improved	73%		
Temporal effects	Onset	30-60 mins	53%		
		4-8 hours	41%		
		8+ hours	41%		
	Non-dosing day	Positive	47%		
		Neutral/Baseline	41%		
Challenging Effects		Yes	91%		

Note. This table presents themes mentioned in at least one-third of survey responses (>33%), with the exception for microdosing regime which was included for context. Percentages represent the proportion of respondents who answered each question.

Demographic Characteristics: Middle-Aged Males Prevalent in Sample

Participants ranged in age from 20 to 61 with a mean age of 40.14 ($SD = 10.77$). Most respondents identified as male (79.17%) and live in the United States of America (33.36%). Regarding microdosing status, 11 participants (40.7%) reported they were currently microdosing at the time of the survey, while 14 participants (51.9%) were not currently microdosing but had done so in the past and intended to microdose again. Only two participants (7.4%) reported having microdosed in the past with no intention of continuing the practice.

Diagnoses: ADHD, Depression, and Anxiety Improvements

While a third of participants reported no medical diagnosis, the majority reported having a psychiatric disorder. Participants with psychiatric conditions typically reported multiple diagnoses ($M = 2.64$, $SD = 1.45$), with ADHD, depression, and anxiety being the most prevalent conditions. When asked about symptom changes since beginning microdosing, most participants reported improvements across their psychiatric conditions. Notably, every participant who reported a change in depression symptoms indicated

improvement (e.g., “Single handedly lifted me out of depression when nothing else worked (SSRI, SNRI, psychotherapy...)”). Similarly, most individuals who reported a change in anxiety symptoms mentioned an improvement (e.g., “Ruminating thoughts stopped while on it, thank god. It did improve quite a bit”). In contrast, for ADHD, the majority of individuals who reported a change stated that microdosing had no effect on their symptoms (e.g., “Only ritalin helps this for me - Little change”).

Additional Substances: Caffeine, Cannabis, Alcohol, and Conventional Psychedelics

Common

The majority of participants reported using various substances while microdosing. Caffeine was the most commonly mentioned substance, followed by cannabis, alcohol, nicotine, and macrodoses of psychedelics (both psilocybin and LSD). In addition, a large proportion also reported taking medication, which was most commonly used to treat ADHD (including Vyvanse/Elvanse, Adderall, Dextroamphetamine/Dexedrine, and Ritalin). When asked about their frequency of use, participants typically reported daily or almost daily consumption of caffeine, nicotine, and medications, while macrodoses of psychedelics were generally consumed infrequently (once or twice).

Microdosing Approaches

Substance: Psilocybin Dominates as the Preferred Choice

LSD and Psilocybin were the only two substances reported in our survey. Psilocybin was the most frequently used substance, with doses ranging from as low as 90mg to as high as 600mg ($M = 260.58\text{mg}$, $SD = 175.19$). One participant reported microdosing psilocybin truffles rather than mushrooms. This was included in the overall substance frequency statistics for psilocybin but excluded from dosage calculations due to potential potency

differences. LSD was used by approximately a third the sample, with doses ranging from 6ug to 50ug ($M = 16.42\text{ug}$, $SD = 16.58$).

Regime: Three to Four Weekly Doses Most Common

Regarding regime, the most frequently reported microdosing schedule was three to four times per week. This includes participants who specifically report microdosing three or four times per week, as well as those that report every second day, and three days on, two days off. The most popular specific regime reported in our survey was dosing every third day.

Additional Practices: Exercise and Meditation Frequently Incorporated

Seven participants did not respond to the question asking about additional practices in their microdosing routine and were thus removed from analysis. Among those who responded, the majority reported including exercise (e.g., “make sure I take a walk”) and meditation as part of their microdosing routine.

Decision Factors: Accessibility and Self-Experimentation Guide Choices

Participants were asked “How did you decide on each of the following components of your microdose?” alongside a microdosing component. Unfortunately, five participants misinterpreted the question regarding substance, and six misinterpreted it for dose, and were thus removed from analysis. The most common reason for deciding on a substance to microdose was due to the ease of access (e.g., “Psilocybin was available from a co-worker”). In regard to selecting a dose, the most frequently reported way of deciding on dose was through trial and error (e.g., “Trial and error to find a barely perceptual dose”).

Duration: Most Users Continue Microdosing Beyond Short-Term Experimentation

Participants were asked to indicate how long they had been microdosing psychedelics. They reported microdosing for periods as short as three weeks to as long as ten years with the average duration being 1.79 years ($SD = 2.72$). Participants frequently reported either

microdosing for longer than six months or for a period between one to six months, indicating a tendency toward sustained rather than brief experimentation.

Intentions: Driven by Positive Psychedelic Experiences and Health Goals

Participants were asked to describe what led them to start microdosing, and what were their expectations. Three main themes were extracted from the responses. First, the most frequently reported motivation for microdosing was to improve their mental health (58%) (e.g., “I was expecting an improvement in my mood. And reduced anxiety”). Second, participants often reported microdosing because of a positive experience they had previously had with psychedelics (42%) (e.g., “I have macrodosed multiple psychedelics in the past and missed the beneficial aspects”). Third, a noticeable proportion reported microdosing to improve their cognitive functioning (32%) (e.g., “my main focus was boosting creativity and focus”).

Temporal Effects: Onset Within an Hour with Daylong Duration

Participants were asked to indicate when they typically begin to notice the effects of their microdose and how long these effects typically last. Participants report noticing the effects of a microdose ranging from immediately, up to two to three hours later, with most beginning to notice the effects between 30 to 60 mins after dosing. They reported experiencing these effects for as short as a few hours, to as long as a couple days, with an average of 11.12 hours ($SD = 11.47$). The most frequently reported durations of effects was split between four to eight hours and periods exceeding eight hours. Participants were also asked to report what they experience on non-dosing days. The majority reported continued positive effects on non-dosing days, although half of these positive reports state that the effect is not as strong or diminishes over time (e.g., “Improved mood carried over from the day before. Not as strong obviously but still much better than pre microdosing days”). A

large number of participants also reported regressing back to their baseline state on non-dosing days (e.g., “back to normal”), indicating that for many, the effects are only temporary.

Cognitive Changes: Substantial Improvements Across Multiple Domains

Participants were asked to describe how they viewed their mental performance in the following five cognitive domains, before and after they began microdosing: focus, memory and learning, problem solving and creativity, clarity of thought and clear headedness, and organisation skills. Prior to microdosing, most participants described their mental performance as poor in focus, organisation skills, and clarity of thought. In contrast, their performance was frequently described as good in problem solving and creativity, and memory and learning. After microdosing, many participants reported an improvement in all cognitive domains, with clarity of thought and focus showing the most frequently reported improvement, followed closely by problem solving and creativity. Reported improvements in memory and learning, and organisation skills were less frequent with a considerable proportion also reporting no effect of microdosing on these domains. Table 10 displays representative comments for each of the frequent themes prior to microdosing and after microdosing.

Table 10:

Representative Comments for Pre and Post Cognitive Abilities

Prior cognitive abilities	Direction	Example comments
Focus	Poor	“Bad, easily distracted, prone to procrastination”
Organisation	Poor	“Almost non existent, dumpster fire”
Clarity of thought & clear-headedness	Poor	“I often have 3-5 different trains of thought going on at any one time. This can be stressful and cause anxiety.”
Memory and learning	Good	“I have a high IQ (am in Mensa)”
Problem solving and creativity	Good	“I'm a talented artist, naturally creative in many different ways, and always have been since I was very little.”
Post cognitive abilities		
Focus	Improved	“Yes, noticeably enhanced focus”

Prior cognitive abilities	Direction	Example comments
Memory and learning	Improved	“Recollection was boosted, I don't know if anything new was learned, but I could make connections between ideas with a bit more ease.”
	No effect	“No noticeable impact”
Problem solving and creativity	Improved	“I am more creatively inspired, more moved and awed by music and art and nature.”
Clarity of thought & clear-headedness	Improved	“Because the default mode network was switched off, it was incredible. I had transformed from a guy locked in his mind into somebody living in the present, who can think clearly”
Organisation skills	Improved	“Improved, I've organised my office and home to be more manageable”
	No effect	“No change most of the time”

Broader Changes: Widespread Improvements in Mood, Social Functioning, and Outlook

Participants were asked to describe any changes they experienced across four domains since microdosing: affective, social, physiological, and insights/transformations. Most participants reported benefits in the affective domain, with the majority stating positive mood changes. Similarly, social enhancements were frequently reported, as well as improvements in insights/transformations, measured by an improved outlook and better habits and lifestyle changes. Physiological changes were measured across five sub-questions, investigating changes in energy, sleep, senses, athletic/physical performance, and physical/bodily sensations. The majority of participants reported improvements in energy levels, sleep, and sensory changes, while stating no effect of microdosing on athletic/physical performance, or on physical/bodily sensations. Table 11 reports representative comments for the frequent themes.

Table 11:*Representative Comments for Emotional, Physiological, Social, and Insight/Transformation**Changes*

Theme	Direction	Example comments
Emotional changes	Improved	“Yes. I am happier, calmer, laugh more easily, less emotionally dysregulated, more grateful, more moved by music and art, more connected with others. Calmer and happier and more of my true self. Trauma symptoms are less torturous.”
Physiological changes		
<i>Energy changes</i>	Improved	“Yes, elevated energy and arousal”
<i>Sleep changes</i>	Improved	“Improvement. Better sleep. More refreshing.”
<i>Sensory changes</i>	Improved	“Sight wholly improved. I have 20/20 vision, but when I am depressed, my vision turns from clear into almost a very slight bit of fog, like the faintest TV raster. Hearing, I felt like my normal tinnitus was more easily ignorable. Smell was drastically improved, could actually smell my dinner. I felt like my nostrils opened up when usually they feel stuffy. Taste, not sure, probably improved. Touch was improved. Difficult to describe but I felt like I could more easily make out details by touch alone. It was easier for me to feel which string of the guitar I was playing without looking.”
	No effect	“None”
<i>Athletic/physical performance</i>	No effect	“No change”
<i>Physical sensations</i>	No effect	“No change”
Social changes	Improved	“Yes. More comfortable socially and more engaged and interested in others and greater ease and more positive and more “present” when in social situations or speaking with and or around others overall, not just friends and family, but overall.”
Insights and transformations changes		
<i>Habit/lifestyle changes</i>	Improved	“I try to engage more with my surroundings as if I am microdosing. Like, I try to bring myself into that same mindset where I could be awestruck by ordinary things around me at any given moment. I find that to be a very healthy practice (and a very easy thing to do when you live in New Zealand).”
<i>Outlook changes</i>	Improved	“I am less negative and I have less suicidal thoughts. Which were very present to the point of paralysing my life.”

Unwanted Effects: High Doses Associated with Fatigue and Irritability

All but one participant reported experiencing an unwanted or challenging effect from microdosing. Of the ten participants who reported a negative experience from microdosing, there were two common themes that emerged. These were challenges regarding fatigue (e.g., “sometimes I can’t keep awake”) and irritability (e.g., “I didn’t notice at first but it was easy for me to lose my temper.”). When asked what they thought contributed to these unwanted experiences, the most common response was that their dose was too high (e.g., “I actually completely screwed up my microdosing method. Because I was a rookie, didn’t really know what I was doing and simply cut the tab in half and consumed it. I actually failed to realise later that if a tab is about 100ug, I would have been theoretically taking 50ug instead of the recommended 25. So it wasn’t actually a microdose”). This suggests that tailored dosing is critical for minimising adverse effects during microdosing.

Chapter Four: Discussion

Our study employed a double mixed methods design, combining a qualitative and quantitative content analysis of 101 posts from the r/microdosing subreddit and an exploratory survey of 28 participants. This approach allowed us to investigate the effects of microdosing on both cognitive functioning, and potential mediating factors. Our findings demonstrate frequent reports of cognitive enhancement, particularly in attentional processes and cognitive clarity, with moderate improvement in higher-order cognition and memory/learning. These cognitive benefits occurred alongside widespread improvements in mood, energy levels, and social functioning, suggesting microdosing may function as a multi-domain enhancer rather than a targeted cognitive intervention. The following discussion examines who engaged in microdosing, the microdosing practices they followed, and the positive and negative effects they report. We then explore the implications of our findings for current psychedelic research models and address the methodological limitations of our approach. Finally, we suggest directions for future research that may help bridge the gap between naturalistic reports and laboratory findings.

Who Engages in Microdosing

Demographic Patterns: Age and Gender

The sample used in our study predominantly consisted of males with an average age of 40. The larger percentage of males in our study aligns with existing microdosing literature, which consistently reports higher proportions of male participants (as seen in, Murphy, Muthukumaraswamy, et al., 2024; Rosenbaum et al., 2020). However, our sample was noticeably older than previous research. Lab-based studies typically recruit participants between the ages of 18 and 35 years, and naturalistic studies report mean ages ranging from 27 to 35 (as seen in, Murphy, Muthukumaraswamy, et al., 2024; Rosenbaum et al., 2020). Unlike other social media platforms, Reddit users are not required to include identifying

information on their profiles, making demographic data collection dependent solely on details voluntarily mentioned within posts. As a result, demographic information is typically neglected in studies analysing passively collected Reddit data (e.g., Lea et al., 2020; Miura, 2022). This pattern is confirmed by the low frequency counts in our own content analysis where only 20 users reported their age and 14 users reported their gender. Nevertheless, the demographic information captured in our content analysis follows the same pattern observed in our survey data, indicating consistency across data sources. It is possible that this age difference may reflect the demographics of the r/microdosing subreddit, or it could potentially indicate an interest among older adults to seek cognitive enhancement or relief from age-related cognitive changes.

Prevalence of Psychiatric Conditions

Another notable demographic feature was the high prevalence of reported psychiatric disorders among participants. Approximately half of the participants in our study reported experiencing a psychiatric disorder, with depression, anxiety and ADHD being the most common. This finding aligns with existing naturalistic microdosing studies that use data from online communities (Haijen et al., 2024; Hutten et al., 2019; Lea, Amada, Jungaberle, et al., 2020a, 2020b), yet contrasts with experimental microdosing research which typically excludes individuals with psychiatric conditions (As seen in, Murphy, Muthukumaraswamy, et al., 2024). It should be noted that because our primary data was collected from self-reported Reddit posts, we cannot determine whether reported psychiatric conditions were professionally diagnosed or self-diagnosed. As a result, the proportion of psychiatric disorders reported in the reddit analysis may be over represented.

Experimental studies that exclude individuals with psychiatric conditions represents a significant limitation in microdosing research, as it fails to represent real-world microdosers.

This suggests that researchers are prioritising internal validity with homogeneous healthy samples at the expense of external validity. By overlooking the psychiatric conditions frequently reported in naturalistic data, researchers may be reducing their ability to detect the cognitive improvements reported in naturalistic studies. This is because they cannot capture cognitive enhancements mediated by improvements in psychiatric symptoms. Future research would benefit from including both ‘healthy’ participants and those with psychiatric disorder to better understand how psychiatric disorders might moderate cognitive outcomes.

The Role of Pre-existing Cognitive Function

Our findings highlight an important pattern regarding participants’ baseline cognitive functioning. Most participants’ perceived themselves to have pre-existing cognitive impairments before they began their microdosing practice. Specifically, over half of the participants indicated they had poor focus, clarity of thought, and organisational skills, prior to microdosing. This suggests that many individuals may turn to microdosing specifically to address these perceived cognitive deficits. This pattern highlights a limitation in controlled studies that, while measuring individual baseline cognitive performance, typically fail to contextualise whether participants began with below-average, average, or above-average cognitive abilities relative to the general population (e.g., Cavanna et al., 2022; Fallon, 2021). Future research would benefit from including participants with perceived cognitive deficits and those with typical cognitive functioning, to better understand the effect of different baseline abilities.

How do Users Microdose

Substance Selection and Dosage Considerations

Psilocybin mushrooms were the most commonly microdosed substance in our study, being used by approximately two-thirds of our sample, typically in doses less than 200mg.

LSD was the second most frequently reported substance, used by approximately one-third of participants. This contrasts with the distribution in existing literature, where LSD is consistently reported as the predominant substance in both experimental and observational studies (As seen in, Murphy, Muthukumaraswamy, et al., 2024; Rosenbaum et al., 2020). There is limited research specifically examining psilocybin's effects on cognition as only three studies have objectively measured cognitive changes using a psilocybin microdose. In the first study participants selected their own dose, resulting in an average of 200mg (Szigeti et al., 2021), which aligns with the average dose observed in our study. In the second study, participants were administered 500mg (Cavanna et al., 2022), more than twice the average dose reported in our sample. In the third study, participants were given psilocybin-containing truffles, making dose comparisons difficult due to differences in potency between truffles and mushrooms (Prochazkova et al., 2018). The disproportionate use of psilocybin in our sample compared to experimental research may partially explain the positive cognitive effects observed in our study. This is because LSD and psilocybin have different pharmacological effects (Rickli et al., 2016). Overall, this highlights that controlled studies are not adequately reflecting real-world microdosing practices regarding both substance choice and dosage. This difference between laboratory protocols and real-world practices limits the generalisability of experimental findings and may explain the variability in outcomes. Future research should address this by prioritising psilocybin studies at doses consistent with community practices.

Our survey provided the unique advantage of investigating participants' reasoning for selecting substance and dose. We identified that substance choice was influenced by how easy it is to access, highlighting the practical considerations guiding microdosing decisions. More importantly, we identified that respondents' decision on dosage was guided through trial and error. This is particularly interesting as it highlights the experimental and individualised nature of microdosing. This aligns with previous research that has likened

microdosers to scientists, noting how individuals approach microdosing in a regimented and scientific manner (Ryan et al., 2023). Ryan et al., (2023) identified that microdosers commonly experiment with their microdosing practice such as substance, dosage, and regime to determine what works best for their individual needs. This raises concerns about current lab-based studies which fail to replicate the personalised approach seen in naturalistic settings. As current studies typically administer a universal standardised dose, they may be reducing their ability to detect cognitive benefits that only emerge when doses are tailored to the individual.

Dosing Schedules and Frequency Patterns

Dosing schedule is a key component of microdosing that displayed considerable variation in our study, ranging from as little as once a week, to as often as every day. The majority of participants report microdosing three to four times per week, with the most frequently followed schedule being every third day. This aligns with findings from existing naturalistic microdosing studies and reflects Fadiman's influence through his recommendation of the every third day schedule (Johnstad, 2018). However, established dosing schedules are not adequately incorporated into lab-based studies, with most providing participants with a single dose before measuring the effects (e.g., Cavanna et al., 2022; Fallon, 2021). Among the already limited experimental literature, five studies administered multiple doses, and only three implemented the three to four doses per week commonly observed in naturalistic settings. Therefore, to better reflect real-world practices and capture potential cumulative effects, future microdosing research would benefit from incorporating more ecologically valid dosing schedules, such as the commonly observed three to four doses per week.

Stacking: Combining Substances for Enhanced Effects

A significant proportion of participants in our study reported combining their microdose with additional supplements, a practice commonly referred to as 'stacking' within the microdosing community (Pop et al., 2024b). In our study, lions mane mushrooms was the most commonly stacked supplement which aligns with the existing microdosing literature (Rootman et al., 2021). The prominence of lion's mane mushrooms likely stems from research highlighting the potential cognitive enhancing properties and the potential for a complementary effect with psilocybin mushrooms for improving cognitive function (Mori et al., 2009). As the beneficial properties of both lions mane and psilocybin are proposed to operate via BDNF-related processes, an additive effect may partially explain the cognitive improvements frequently reported in naturalistic settings (Vigna et al., 2019). Therefore, future research should aim to distinguish which effects are caused by psilocybin and which are caused by lions mane.

Positive effects

The most notable finding from our analysis was the overwhelmingly positive sentiment expressed by participants regarding their microdosing experiences. In our content analysis, most Reddit posts received a sentiment score of four or five on a five-point scale (85%), indicating positive attitudes toward microdosing. These positive attitudes extended across multiple domains, reiterating the belief that microdosing is a multi-enhancer that produces benefits spanning various areas of well-being. The following section will detail the positive effects across cognitive, affective, physiological, and social domains, and how these factors may contribute to participants' cognitive outcomes.

Cognitive Enhancements

Addressing our primary research question, participants consistently reported improvements in cognitive functioning related to their microdosing practice. The most frequently reported domains of cognitive enhancement were attentional processes and cognitive clarity. Our findings regarding attentional processes present a notably more positive picture that contrasts with lab-based research, which predominantly reports either no effect or impairment when using objective measures. However, our results align with existing naturalistic studies, which typically report improvements in attention (e.g., Anderson et al., 2019; Cameron et al., 2020; Lea et al., 2020; Petranker et al., 2022). Interestingly, cognitive clarity, despite being a common finding in our study, is largely neglected in existing literature, with no lab-based studies measuring this domain (e.g., Murphy et al., 2024). Even among naturalistic studies, only two acknowledged mental clarity as a distinct cognitive domain, though when mentioned, the improvements were consistent with our findings (Fernandes-Nascimento et al., 2022; Petranker et al., 2022). The frequency of reported improvements in cognitive clarity in our study may reflect relief from the long-term neurological consequences of the recent COVID-19 pandemic. Particularly the subjective experience of 'brain fog', where patients commonly describe feelings of 'fuzziness' and 'blur' in their thinking processes (Sia et al., 2023). Therefore, mental clarity may represent an important domain of cognition that traditional assessments fail to capture, and may be linked to attentional improvements, as mental clarity is likely to aid in sustained focus.

Improvements were also reported in higher order cognition and memory/learning domains, though these enhancements were less universally experienced than those for attention and clarity. Regarding higher order cognition, participants in our study who discussed creativity almost invariably reported improvements, though the overall frequency of such mentions was relatively low. This pattern could suggest that creativity improvements

were less universal, with the lack of mention potentially implying no noticeable change in the domain. If so, this finding matches the mixed literature on creativity, where experimental studies show mixed results, but naturalistic research show consistent improvements. Similarly, our results indicate that improvements in memory and learning are less frequently discussed than the domains described above, although still demonstrate a pattern towards improvement if mentioned. This finding portrays a slightly more positive picture on memory that contrasts with the existing experimental and naturalistic research that frequently identify no effect on memory. Overall, our research suggests that microdosing produces more consistent effects on attentional processes and cognitive clarity than on higher order cognition and memory/learning, even though laboratory research has focused primarily on the latter domains.

This pattern of cognitive effects aligns with our initial hypotheses, which predicted microdosing would produce a prominent improvement in prefrontal cognitive functioning, and that regions with higher densities of 5-HT_{2A} receptors, such as the frontal cortex (involved in attentional processes and executive functioning), would show greater improvements compared to regions with lower receptor densities, such as the hippocampus (involved in memory and learning). The more pronounced improvements in attention processes compared to memory and learning observed in our study support this neurobiological framework. Suggesting that the distribution of 5-HT_{2A} receptors may influence which cognitive domains are most affected by microdosing.

Interestingly, the cognitive domains showing the greatest improvements, focus and clarity of thought, were also two domains where participants frequently reported pre-existing deficits. This pattern suggests microdosing may be particularly effective at remedying pre-existing cognitive impairments, rather than enhancing already adequate cognitive abilities.

This finding could help explain discrepancies between lab-based and naturalistic research. Lab-based studies typically recruit ‘healthy’ individuals but don’t necessarily screen for subjective cognitive deficits. This may limit observable improvements due to ceiling effects if participants begin with a relatively good baseline cognitive functioning. The strong relationship between pre-existing impairments and subsequent improvements suggest future research would benefit from separating participants based on baseline cognitive performance. This approach may provide a deeper understanding into which populations might benefit the most from microdosing.

Affective Improvements

Affective changes emerged as one of the most frequently reported domains of improvement in our study, with over half of the participants describing enhanced general mood states since microdosing. In combination with these affective improvements, a substantial proportion of our sample also reported improvement in mood disorders (anxiety and depression). These improvements align with the collective findings from naturalistic research, where positive affective changes are overwhelmingly reported. Across these naturalistic studies, participants report reductions in depression and anxiety, as well as general improvements in mood and well-being (e.g., Haijen et al., 2024; Lea et al., 2020; Petranker et al., 2022). In contrast, lab-based studies that administer validated scales and questionnaires have produced mixed results, with several controlled trials failing to detect significant affective improvements while others have identified modest benefits (see Murphy et al., 2024 for review). It is important to note that the only identified worsening of affect in lab-based studies are transient increases in anxiety, and this was observed exclusively in individuals that received higher doses (26ug), with no anxiety effects reported at lower doses (6.5ug, 13ug) (Bershad et al., 2019). Our findings contribute to the ongoing discussion

regarding discrepancies between naturalistic and controlled studies, by reinforcing that microdosers consistently report affective improvements in real-world contexts.

The widespread affective improvements reported in our study may potentially mediate the observed cognitive enhancements. From a psychological perspective, affective improvements could influence how one perceives their cognitive abilities through a phenomenon similar to ‘positive cognitive bias’ where individuals create their own “subjective reality”. When individuals experience relief from mood disorders or enhanced mood states, they may develop more positive self-evaluations on cognitive abilities. This is supported by research that demonstrates depressed individuals report poorer cognitive functioning compared to healthy individuals, even when objective performance is comparable (Szyszkowska & Bala, 2023). In addition, prior research has also concluded that positive affect has the potential to improve cognitive functioning (Pupillo et al., 2022). Overall, the high prevalence of psychiatric conditions in our sample and other naturalistic studies is noteworthy when considered alongside these reported improvements. The mutual enhancement in both affective states and cognitive functioning highlights a potential moderating relationship between improved psychiatric symptoms and cognitive benefits. Therefore, improvements in cognitive functioning often reported in naturalistic settings might partially be accounted for through improvements in emotion and psychiatric disorders.

Positive Lifestyle and Perspective Changes

Microdosing often coincides with positive lifestyle changes and perspective shifts. Based on our observations of our Reddit data, participants appeared to begin microdosing as part of a broader transition toward healthier living. This overlap between positive lifestyle changes and microdosing makes it difficult to distinguish which benefits resulted from microdosing and which occurred from other lifestyle changes. This is because improvements

in cognition are independently associated with commonly reported lifestyle changes, such as reducing substance use and increasing exercise habits (Gomez-Pinilla & Hillman, 2013; see Sampedro-Piquero et al., 2019). Similarly, many participants also described a more positive view of themselves and on life in general. This improved perspective may also influence how an individual evaluates their cognitive abilities. This is supported by de Witt and colleagues (2022) who found participants reported greater performance on cognitive tasks after microdosing, despite displaying comparable performance. These interconnected changes suggest that cognitive improvements may be mediated through this broader context of lifestyle enhancements and shifts in perspective rather than a purely pharmacological effect of microdosing.

This pattern has important implications for current lab-based studies, as many use much shorter microdosing durations than observed in real-world settings. The short timeframes in lab-based research likely prevents researchers from observing cognitive improvements that develop gradually and miss the additive effects of co-occurring lifestyle changes. For example, our study identified frequent reports of reduced substance use as a positive lifestyle change. This is particularly relevant since research has concluded that the cognitive recovery time from substance abuse can take weeks to years (Goldman, 1995). As these lifestyle improvements take time to influence cognition, lab-based studies that only administer a single dose may be failing to capture the potential benefits of prolonged microdosing. Future research should incorporate longer microdosing protocols to better reflect the timeframes identified in real-world practices, and to capture the additive effect of co-occurring lifestyle changes.

Positive Physiological Changes

Our study found an interesting pattern regarding energy levels among microdosers. While the majority of participants reported increased energy when microdosing, a small subset in both our Reddit and survey data reported drowsiness as an adverse effect. Similarly, both naturalistic and lab-based studies that administer validated scales generally demonstrate increased energy levels after microdosing (see Murphy et al., 2024; Petranker et al., 2022). Notably, a few naturalistic studies also acknowledge that a small proportion of participants also experience fatigue or drowsiness, thus matching the contradictory patterns observed in existing microdosing research (e.g., Petranker et al., 2022). This variability in energy responses may partly explain the mixed cognitive outcomes observed in microdosing research. Given the relationship between decreased energy levels and impairments in cognitive performance, the physiological effects of microdosing on energy states should be carefully considered when investigating cognitive outcomes (Neu et al., 2011). Future research would benefit from examining whether cognitive improvements correlate with energy increases, and whether reported cognitive impairments coincide with experiences of drowsiness, potentially revealing an important mediating pathway.

Improved Social Functioning

Our findings also highlight improvements in social functioning following microdosing, with nearly three-quarters of survey respondents reporting enhanced sociability. Interestingly, unlike other domains where specific subthemes were frequently captured, social improvements appeared more diffuse, spanning across multiple areas without a single aspect dominating. This pattern suggests that microdosing may not target specific social capabilities but rather produces a broad enhancement across various dimensions of social functioning. This aligns with the findings in naturalistic research which frequently reports widespread social improvements (e.g., Anderson et al., 2019; Lea et al., 2020). However, lab-based

studies using validated measures are less consistent, typically reporting mixed effects with a tendency towards social benefits (E.g. Szigeti et al., 2021; Molla et al., 2024; Hutten et al., 2020). This discrepancy may reflect limitations in lab-based settings, which often lack the social contexts where improvements can naturally occur.

Negative Effects

Despite the overwhelmingly positive results observed in our study, it is important to highlight that microdosing is not a risk-free practice. Although rarely identified in Reddit posts, our survey found that 10 out of 11 respondents reported experiencing challenging or unwanted effects from microdosing. Irritability and drowsiness were two common challenges identified, which aligns with the challenging effects identified in existing naturalistic research (as seen in, Anderson, Petranker, Christopher, et al., 2019; Lea, Amada, Jungaberle, Shecke, & Klein, 2020). Interestingly, this contrasts with the pattern towards increased emotional regulation and increased energy levels observed in our study. When asked what participants thought led to unwanted effects, a common theme was problems with dosage. This emphasises the need for personalised doses when microdosing, and highlights there may be a fine line between practices that can cause enhancement and practices that can cause impairments.

Investigating temporal effects in our survey highlights another important consideration regarding microdosing practices. When asked to describe their experiences on non-dosing days, a complex pattern emerged. While many reported that positive effects remained following their dosing day, they often noted that the benefits diminished over time, highlighting the temporary nature of positive effects. This observation raises concerns about dependency, as individuals may feel compelled to continue microdosing to maintain their desired cognitive and emotional state. One participant explicitly described how their life

began to follow a “three day chapter” around their microdosing schedule. They reported increased energy and confidence on dosing days, followed by feelings of drowsiness and everyday challenges on day two. Returning to normal on day three, with the process repeating again with a dose on day four. This suggests that rather than producing sustained improvements, microdosing may create a cycle of enhancement and impairment that could potentially lead to psychological dependence. This finding highlights the need for future research to formally investigate the effects of long-term microdosing and whether there is a need for staggered long-term breaks within the practices.

Implications

The findings from our study have important implications for the current model of psychedelic research, highlighting a disconnect between naturalistic microdosing practices and experimental lab-based studies. While traditional research follows a standardised approach, administering uniform doses on fixed schedules, our data support previous research that highlights microdosing is an individualised practice. Participants consistently described experimenting with different dosages and schedules, with many stating that trial and error was central to finding their optimal dose. This experimental approach contrasts with lab-based studies that administer standardised doses without accounting for individual variability or allowing for dose adjustment. This suggests that current research methodologies in the microdosing literature may be following an outdated way of researching these substances. To better replicate the improvements seen in naturalistic studies, microdosing research needs to implement more flexible protocols to accommodate for the individualised approaches seen in real-world contexts. The rigid pharmaceutical models currently used fail to capture the personalisation of the practice, which is a core feature. Without this methodological shift, clinical studies risk continuing to produce results that inadequately reflect the experiences reported by real-world microdosers in naturalistic environments.

Further, current psychedelic research on macrodosing often emphasises the importance of mystical experiences as the primary mechanism underlying therapeutic outcomes for conditions like depression, addiction, and anxiety disorders (Romeo et al., 2025). Mystical experiences are suggested to play a crucial role in the therapeutic effects of psychedelic therapy because it causes an abrupt, substantial, and sustained change in behaviour and perception (see Ko et al., 2022). If future microdosing research can confirm the positive cognitive effects reported in naturalistic settings using objective measures, this would challenge the current view in psychedelic research that mystical-type experiences are necessary for therapeutic benefits.

Limitations

The use of naturalistic data in our study provides better external validity than experimental lab-based studies, as it provides an accurate view of how psychedelics are microdosed in a real-world setting. Our study has identified that different data sources yield similar yet slightly different results, further supporting the need for diverse data sources in this kind of research. Although using naturalistic data does not come without its limitations that cannot be controlled for due to its observational nature.

Our study utilised Academic Torrents to extract data from the r/microdosing subreddit, which presents potential limitations that should be acknowledged. The need for Academic Torrents arose as a result of recent restrictions to the official Reddit API which has been used by previous studies (see Pop et al., 2024). A torrent is a peer-to-peer file sharing system where multiple users host and share copies of the same dataset, unlike typical data collection methods that is extracted from an original source. This raises concerns about the use of peer-to-peer sharing for academic research where collection is dependent on having users who are able to seed the data, potentially introducing inconsistencies in data reliability.

Therefore, unlike official databases, there is no guarantee of data completeness. Additionally, the use of torrented Reddit data raises ethical considerations about consent, as posts that have been deleted or edited after the initial data collection are not reflected in the torrented data.

Another limitation of the Reddit data set stems from the unprompted nature of individuals' posts. Posts were given an information score from one to five, which assessed how well the post described microdosing effects across various domains. The majority of posts received relatively low information ratings (between one and three), indicating limited detail about the effects of microdosing. This highlights the importance of our supplementary survey, which allowed us to ask targeted questions about cognitive effects and potential mediating factors lacking in spontaneous Reddit discussions.

It's important to acknowledge that our study captures only a subset of the microdosing population, specifically, those who use Reddit and are motivated to discuss their experiences publicly. This may introduce self-selection bias and skew our data toward extremes, as users are typically more motivated to report notably positive or negative experiences rather than subtle or ambiguous effects. Furthermore, posts were selected for manual coding based on their upvote-to-downvote ratio. This approach may have prioritised posts that aligned with the community's positive expectations on microdosing, as users are more likely to upvote content describing successful experiences rather than disappointing or negative outcomes. Overall, our findings may present an overly optimistic view of the cognitive effects of microdosing compared to the full spectrum of user experiences.

Although the use of Reddit has been acknowledged in the scientific literature as a valuable data source, there are limitations that cannot be addressed due to its observational nature. First, as psychedelics are illegal in many places of the world, there is no guarantee the substances used are pure. This can be supported by the low testing rates typically associated

with psychedelics (Petranker et al., 2022). This concern is also acknowledged by a doctor who stated that one individual thought they were consuming psilocybin but chemical analysis revealed a combination of different drugs (Miller, 2022). Second, psychedelics lose potency over time depending on storage conditions such as exposure to light and heat (Gotvaldová et al., 2021). This degradation introduces variability in the dose consumed which cannot be addressed in observational studies as storage conditions are unknown. Therefore, participants may be unknowingly taking much lower doses than reported. These uncontrollable variables significantly affect the accuracy of findings in observational psychedelic research.

Another limitation of analysing Reddit data is the inherent ambiguity regarding users' experiences. When participants discussed conditions such as depression and anxiety, it was often difficult to differentiate between those diagnosed by a medical professional and those who are self-diagnosed. This ambiguity potentially conflates the number of clinical disorders with temporary emotional states, making it difficult to draw reliable conclusions about the therapeutic effects of microdosing. Similarly, the temporal component of reported effects was often difficult to distinguish. This made it challenging to determine whether reported cognitive changes occurred acutely following administration, emerged gradually throughout the day, or appeared during non-dosing days as residual effects.

A notable observation from our study is the difference in microdosing duration between our two data sources. Survey respondents were predominantly experienced users who had over a year of experience, while Reddit users typically reported durations of less than a month. This difference likely reflects the different roles these platforms serve in the microdosing community. Reddit appears to attract those new to microdosing who are seeking advice during their early experiences (Pop et al., 2024a). In contrast, our survey seems to have appealed to experienced microdosers who feel more confident discussing their

experiences and may have developed more stable perspectives on the practice. This observation has important methodological implications for future research, as sampling exclusively from either population could introduce significant bias. Studies drawing primarily from Reddit may overrepresent newer users still navigating initial effects, while surveys tend to capture experienced users who continued microdosing because of positive outcomes. This highlights the need for future researchers to consider where they sample their participants from when designing naturalistic studies or interpreting existing literature on microdosing experiences.

To mitigate limitations in the Reddit data, we administered an exploratory survey that allowed us to ask directed questions about aspects of microdosing often neglected in unprompted Reddit posts. While this approach strengthened our methodology, our survey had its own limitations. Most notably, we obtained a relatively small sample size with high dropout rates, with only 11 of 29 participants completing the entire survey. Although this limits generalisability, the comprehensiveness of questions provided in depth detail that partially compensates for the limited sample size. The high dropout rate, while concerning, was anticipated given the survey's extensive nature, a necessary trade-off to gather detailed information regarding the complex relationship between microdosing and cognition.

Future Directions

Investigating Adverse Effects

As discussed, the findings from our study show a predominantly positive picture of microdosing across multiple domains, consistent with naturalistic studies. However, it is important to note that most participants in our survey also reported experiencing adverse effects such as fatigue and irritability. Although these adverse effects were generally not severe, they highlight that microdosing is not a risk-free practice. Research has highlighted

potential cardiovascular risk associated with psychedelics at macrodoses, yet this risk has not been investigated in microdosing research. Our study may have introduced biased sampling towards positive effects due to methodological sampling, and therefore, future research should recruit samples who have experienced adverse and null effects to provide a more balanced view. Our study found that dosage related difficulties may be associated with the adverse effects reported in our survey sample, although more rigorous investigation is needed to identify what factors may be predicting these negative effects and to establish clear safety guidelines for microdosing practices.

Advancing Automated Analysis Methods

Our study initially aimed to apply automated coding techniques to the larger dataset of 8,400 posts, following methodologies established by Wightman and colleagues (2023). Utilising machine learning models for automated coding would have significantly expanded our analysis beyond the 101 manually coded posts, providing a more diverse sample of microdosing experiences. However, despite numerous attempts to train supervised machine learning techniques with our manually coded data, we were unable to produce reliable classification results. This methodological challenge highlights the complexity of self-report posts where the cognitive effects discussed are heavily context dependent. Future research should investigate the use of natural language processing specifically developed to address the context dependent nature of cognitive changes. Successfully implementing automated coding for large Reddit datasets would allow for more comprehensive and efficient investigation of complex human experiences without the resource constraints associated with qualitative research.

Mixed Methods Approaches

To address the methodological gap between naturalistic experiences and laboratory studies, future research should adopt a mixed approach that combines experimental freedom with objective measurement. A current citizen science study, microdose.me, displays a promising methodological design that aligns with this suggested approach. Microdose.me is a downloadable app that participants can join at any time. Participants are initially provided with a questionnaire to capture basic demographic information, medical history, a self-reported use of psychedelics. Over a three-months period, participants are presented with short daily questionnaires alongside a monthly cognitive assessment. Overall, this approach allows individuals to adjust their microdosing practice freely while still capturing objective cognitive data. This methodology follows the individualised nature identified in our study and avoids the standardised constraints of experimental research. Further, the three-months period extends beyond the acute effects typically measured in laboratory settings which can help distinguish between immediate, cumulative, and sustained effects. Moving forward, we recommend researchers implement similar methodological approaches that maintain scientific rigor while acknowledging the personalised nature observed in real-world settings.

Potential Applications of Generative AI

Another promising avenue for future research lies in the application of generative AI. Previous research has suggested that Large Language Models (LLMs), trained on vast internet datasets (including microdosing discussions), could generate synthetic data to explore research hypotheses, by simulating participant responses to open-ended questions. LLM's are typically trained on enormous internet data sets such as Common Crawl, which includes a copious amount of online discussion (Raffel et al., 2020; Common Crawl, n.d). This approach offers rapid, cost-effective data generation and could help navigate the legal barriers currently limiting human research with psychedelic substances. While generated data

cannot replace actual human experiences, it could serve as a valuable preliminary research tool when validated against real participant data. Future studies could investigate the effect of contextual factors by finetuning prompts with the help from microdosing communities to create more nuanced simulations of cognitive experiences. As LLM capabilities continue to advance, they may provide a novel approach to investigate the spectrum of cognitive effects reported in microdosing communities before proceeding to controlled human trials, potentially bridging the gap between subjective reports and laboratory findings. It should be noted that generated data is not as valuable as real human participants and LLMs are known to produce biases and factual errors. Therefore, findings should be validated with real human participants as recommended in prior research (Hämäläinen et al., 2023).

Conclusion

To conclude, a quantitative content analysis of the r/microdosing subreddit and an exploratory survey were conducted to investigate the effect of psychedelic microdosing on cognitive functioning, and what factors may be mediating this effect. Cognitive improvements were frequently reported in both data sources with improvements most frequently reported in domains capturing focus and mental clarity. Our findings align with existing naturalistic research and support the view of microdosing as a multi-enhancer with improvements frequently reported across multiple domains. Due to the vast co-occurring improvements observed, it is possible that microdosing may act more as a catalyst than a targeted intervention, initiating a flow on of mutually reinforcing improvements across multiple domains, rather than affecting isolated systems. Future research should investigate the reported negative cognitive effects of microdosing and identify potential mediating factors to develop a more comprehensive understanding of the relationship between microdosing and cognition. Further, to address the disparity between experimental and naturalistic findings, researchers should incorporate more flexible protocols that acknowledge

the individualised nature of microdosing practices seen in real-world contexts. By incorporating the personalised practices, future studies may better capture the cognitive effects consistently reported in naturalistic settings.

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Appendices

Appendix A: Survey Recruitment Post

[Academic Research] Exploring Individual Differences in Microdosing Effects - Survey Participants Needed

Hello [r/microdosing](#) community!

I am a student researcher at Victoria University of Wellington, working under the supervision of Professor Bart Ellenbroek and Dr Jiun Youn. We are conducting a study to better understand the cognitive effects of microdosing psychedelics and why different people experience varying effects.

Why This Research Matters

Current scientific literature shows contradictory findings about microdosing effects on mental performance - some studies report cognitive improvements, others show impairments, and some find no effect at all. We believe these mixed results might be due to individual differences and various factors that influence outcomes. Your experiences can help us uncover these patterns.

Who Can Participate?

- Required to have microdosed psychedelics at least once in your lifetime
- Can read and write in English

What's Involved?

- Online survey about your microdosing experiences
- Questions about demographics, microdosing practices, and effects you've experienced
- *Completely anonymous* - No IP addresses collected
- Skip any questions you prefer not to answer

The survey link is: https://vuw.qualtrics.com/jfe/form/SV_ezVGGtbHQv6IbJg

This research has been approved by the Victoria University of Wellington Ethics Committee (Approval Number: 2024/HE031472).

Questions or Concerns?

If you have any questions about the study, all contact details for the research team can be found at the bottom of the information sheet at the beginning of the survey found [here](#).

Thank you for considering participation in this research! Feel free to share the survey with friends or family who may be interested.

Appendix B: Survey Information Sheet

Microdosing and mental performance: Tell us your experience

Thank you for your interest in taking part in our online survey! Before you begin, here is some information about the study and the data we will be gathering.

Researchers

Student Researcher: Luke Pelham

Supervised By: [Professor Bart Ellenbroek](#) & [Dr Jiun Youn](#)

Who can take part?

To be eligible to take part in this survey, you are required to have microdosed a psychedelic substance at least once in your lifetime.

What is this study about?

Current scientific research shows contradictory findings about the effects of psychedelic microdosing. Studies have reported:

- Improvements in cognitive function (e.g., enhanced creativity, better focus)
- Impairments in cognitive performance (e.g., decreased attention, slower processing)
- No effect

This pattern of contradictory findings suggests that the cognitive effects of microdosing are complex and may depend on various factors and individual differences. Our goal is to explore patterns in outcomes and identify potential mechanisms behind these effects.

About the survey

This survey contains a comprehensive set of questions to help us better understand the variation of effects reported in the microdosing literature. The survey will ask questions that capture:

- Demographic information
- Details about your microdosing practices
- The effects you've experienced from microdosing
- Factors that might influence your experiences

We encourage you to answer as many questions as possible to help build a complete picture of microdosing experiences. You are free to skip any question that you prefer not to answer and continue with the rest of the survey. Every response you provide contributes valuable information to our research.

What do I get out of it?

This study is completely voluntary, and you can choose to exit this survey at any time. Though you won't receive any direct benefits for your participation, your responses are

valuable and will help advance our understanding of the effects of microdosing psychedelic substances.

Is participation anonymous?

Your privacy is fully protected throughout this survey. Although we'll ask for some basic information about you (such as your age and gender), there is no way to link these details back to any individual. We do not collect IP addresses, ensuring complete confidentiality.

What will happen with the information I share?

The findings from this research will be published in a peer-reviewed scientific journal and will be shared on the Reddit communities where many of our participants come from.

Ethical approval:

This study has been approved by the Ethics Committee at Victoria University of Wellington. (Approval Number: 2024/HE031472)

Participant Agreement:

I have read the information provided above. I understand that by clicking the rightward arrow below I am agreeing to participate in this research study. If I choose not to participate, I can close the browser window.

If you have any questions, feel free to contact:

- Luke Pelham (Student researcher) at luke.pelham@vuw.ac.nz
- Professor Bart Ellenbroek (Supervisor) at bart.ellenbroek@vuw.ac.nz
- Dr Jiun Youn (Supervisor) at jiun.youn@vuw.ac.nz

Appendix C: Reddit Frequency Table

Theme	N	Subtheme	N	Level	N	Code	N			
Affective Changes	74	Emotion Regulation	35	Improved	30	Emotional Awareness	9			
						Emotional Control	17			
						Emotional Resilience	8			
						Worsened	10	Emotional Control	9	
				Emotional Resilience	1					
				Mood States	67	Improved		61	Happiness	42
									Relaxed	30
							Sadness (reduced)		7	
		No Effect	1							
		Worsened	9	Anger	1					
				Anger	1					
				Relaxed	7					
				Sadness	3					
				Cognitive Changes	101	Attentional Processes	50	Improved	39	Focus
Task Switching	2									
No Effect	1									
Worsened	15	Focus	1							
Focus		15								
Cognitive Clarity		41	Improved							35
						Thought Organisation	11			
	Worsened					7	Mental Clarity	6		
	Thought Organisation						1			
General Cognition	30	Improved	27				Mental Speed	5		
							Overall Mental Performance	19		
						Productivity	7			
						No Effect	2			
						Worsened	4	Productivity	2	
				Overall Mental Performance	4					
Productivity	1									
Higher Order Cognition	33	Improved	29	Abstract Thinking	1					
				Creative Thinking	17					
				Decision Making	7					
				Problem Solving	6					
				No Effect	2					
				Worsened	3	Creative Thinking	2			
				Abstract thinking		1				
				Decision Making		2				
				Language Processing		17	Improved	17	Articulation	14
					Comprehension				4	
Memory & Learning	21	Improved	18		Information Retention				5	
					Learning Speed				4	
				Long-Term Memory	12					
				Short-Term Memory	5					
				Worsened	3	Information Retention	1			
				Long Term Memory		1				
				Short Term Memory		1				
				Demographics		85	Age	20	20-29	6
22	1									
27	1									
29	1									
Early 20's	1									

Theme	N	Subtheme	N	Level	N	Code	N
Demographics	85	Age	20	20-29	6	Late 20's	1
					6	30+	1
						33	1
						34	2
						Early 30's	1
						Mid 30's	1
				40-49	5	40's	1
						40+	1
						47	1
						48	1
						49	1
						late 40's	1
				50+	3	50's	1
						54	1
						70+	1
		Current Habits	26	Negative Habits	3	Other	1
						Substance use	2
				Positive Habits	23	Meditation	14
						Other	5
						Regular exercise	8
						Therapy	2
		Diagnosis	55	Autoimmune Diseases	1	Arthritis	1
				Neurological Disorders		Autism	4
						Concussion	1
						Dyslexia	1
						TBI	1
						Stroke	1
				Physiological Disorders	1	Tinnitus	2
						Unknown bloating disease	1
				Psychiatric Disorders	48	ADHD	9
						Addiction	1
						Anxiety	22
						Bipolar	3
						CPTSD	5
						Depression	34
						Eating Disorder	1
						OCD	3
						PMDD	1
						PTSD	1
						Mood Issues	1
		Enviroment		5		Negative Enviroment	5
		Family		16		Has Children	9
				Marital Status	14	In Relationship	4
						Married	10
		Gender		14		Female	3
						Male	11
		Occupation		18		Occupation	18
		Prior to Microdosing		56		Has Macro dosed	29
						Medication/Therapy didn't work	14

Theme	N	Subtheme	N	Level	N	Code	N
		Spiritual & Existential Experiences	17	More Intense or Meaningful	17	Life Purpose	1
						Meditation	6
						Nature Connection	7
						Spiritual Connection	3
Microdosing Approaches	100	Additional Substances	26	Medication	3	Medication	3
				Potentially Adictive/ Psychoactive substance	7	Alcohol	1
						CBD	1
						Caffeine	1
						DMT	1
						Nicotine	1
						THC/Marijuana	3
				Supplements	20	Antioxidants	1
						Ashwaganda	1
						Black Pepper Extract	1
						Cocoa	1
						Cordyceps	2
						Creatine	1
						Fish Oil	2
						Green Tea	1
						Lions Mane	13
						Minerals	1
						Niacin	3
						Proteins	1
						Reishi Mushrooms	3
						Saffron	1
						Sulbutiamine	1
						Sulforaphane	1
						Tumeric (Curcumin)	2
						Vitamins	4
						theonine	1
		Dosage	73	LSD Dosage	22	12.1-16ug	7
						16.1-20ug	1
						20.1-25ug	1
						4-8ug	4
						8.1-12ug	8
						Other	1
				Mescaline Dosage	2	Less than 40mg	1
						More than 40mg	1
				Psilocybin Dosage	49	0-100mg	18
						101-200mg	18
						201-300mg	5
						301-400mg	2
						401mg+	4
						Other	2
		Duration	72	Less than 1 month	34	1 week	4
						2nd Dose	1
						3 Days	1
						4 days	1
						6th Dose	1
						9 Days	1
						First Dose	4
						1 Month	10
						2 Weeks	8

Theme	N	Subtheme	N	Level	N	Code	N
				1-6 months	22	3 weeks	3
						2 Months	8
						3 Months	2
						6 Weeks	4
						7 Weeks	1
						3.5 months	1
						4 months	2
						5+ Months	1
						6 Months	1
						Months	2
				6+ months	17	1 year	3
						10 Months	2
						7 Months	2
						8 Months	2
						9 Months	1
						1 year +	1
						18 months	1
						2 Years	3
						3 Years	1
						5 Years	1
		Effects Felt		11		No	2
						Yes	9
		Microdosing Adjustments	32	Changed Dose	26	With Effects	9
						Without effects	18
				Changed Regime	8	With Effects	2
						Without Effects	6
		Pre Conception		7		Fearful	3
						Hopeful	3
						Reluctant	1
						Skeptical	2
		Regimen/ Schedule/ Protocol	55	1-2 Times Per Week	7	2 Times a Week	3
						Every Fourth Day	1
						Once a Week	3
				3-4 Times Per Week	34	2-3 Times a week	3
						3 Time a week	3
						4 Days on	7
						4 Times a Week	1
						Every 2-3 days	2
						Every Few Days	1
						Every Second Day	1
						Every Third Day (Fadimans)	17
				5+ Times Per Week	11	4-5 days a week	1
						5 days on	5
						Every Day	5
				Personal Pattern	3	As Needed	3
		Setting		7		Education	3
						Home	1
						Sport or Recreation	1
						Work	3
		Stopped Microdosing	15	Stopped	8	Effects remain positive	2
						No Effects Mentioned	4
						Some effects Remain	1

Theme	N	Subtheme	N	Level	N	Code	N				
						Went back to baseline	1				
						No Effects Mentioned	3				
						Went back to baseline	3				
						Effects remained	1				
						Substance	86	1P-LSD	4		
							LSD	22			
							Mescaline	2			
						Psilocybin	61	B+ Cubensis	1		
							Blue Magnolia	1			
							Mushrooms				
							Cubensis	6			
							Golden Teacher	5			
							Mushrooms				
							Liberty Caps	1			
							Magic Truffles	1			
							Mckennaii	1			
							Penis Envy	3			
							Psilocybin Unspecified	44			
							Subaeruginosa	1			
						Time of Dose	8	Evening	1		
								Morning	7		
Motives & Intentions	23					Brain Changes	1				
Physiological Changes	56	Energy		29		Improved	26				
						Worsened	8				
		Physical Performance		16		Improved	15				
						Worsened	2				
		Physiological Discomfort	15	Improved	4	Gastrointestinal Problems	1				
						Head-Related	1				
						Other	3				
				Worsened	11	Gastrointestinal Related	3				
						Head-Related	2				
						Other	7				
		Sensory & Perceptual Changes	17	Improved	16	Sight	12				
						Smell	5				
						Sound	7				
						Taste	4				
						Touch	4				
				Worsened	2	Sight	1				
						Touch	1				
		Sleep		6		Dream Related	3				
						Improved Sleep	4				
						Worsened	1				
		Prior Cognition	17	Enhanced	1	General Cognition	1	Intelligent	1		
Impaired	16			Cognitive Clarity	2	Brain fog	1				
						Racing Thoughts	1				

Theme	N	Subtheme	N	Level	N	Code	N
				Higher Order Cognition	1	Not Creative	1
				Attentional Processes	8	Poor Focus	8
				Language Processing	5	Poor Language	5
				Memory & Learning	2	Poor Memory	2
Social Changes	42	Emotional Connection		22		Improved	22
		Interpersonal Skills		19		Improved	17
						Worsened	3
		Social Engagement		11		Decreased	3
						Increased	8
		Social Perception		11		Improved	8
						Worsened	3
Temporal Effects	30	Day of Effect		21		Dosing Day	18
						Non-Dosing Day	14
		Mixed Effects		15		Different Days	8
						Same day Over Time	3
							6
Other	26	External Observation		8		Negative Change	2
						Positive Change	7
		Information Source		21		Informed	9
						Media Content	2
						Online Communities	5
						Personal Networks	4
						Scientific Literature	3
						Uninformed	1
	101	Overall Ratings	101	Informational Rating	1		25
					2		26
					3		26
					4		18
					5		6
				Sentiment Rating	1		4
					2		4
					3		8
					4		64
					5		21